Reputation and Strategic Competition: The Case of an Online Market *

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Abstract

This paper analyses how sellers strategic competition for high valuation buyers in a context resembling on-line markets shapes reputation building incentives, and how it determines the dynamics of prices and reputation itself. Our main result is somewhat counter-intuitive: ceteris paribus, as the reputation of competitors increase (intensity of competition), a sellers incentive to exert effort that helps to success- fully complete a transaction decrease. This result emerges from the interaction between the expected behaviour of future buyers and the evolution of reputations. The "intensity" of competition effect, however, disappears as the number of buyers in the market increases. We then propose an extension to quantify two identification problems related to the reduced-form estimation of the impact of seller reputation measures on revenues, and that have been omitted by recent empirical papers. Biases in the estimation of this impact arise when reputation building and strategic interactions are not controlled for. When sellers compete strategically and build a reputation, the reputation elasticity of revenues is non-monotonic in a seller reputation measures, which, in equilibrium, is correlated with her competitors reputation.

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1 Introduction

Many markets, especially on-line markets such as eBay, Amazon or on-line house renting platforms, are characterized by (at least) two key informational features. First, there is a potential role for reputation building: buyers have little knowledge about the sellers or the quality of a product they are interested in purchasing. Feedbacks and comments by previous buyers, as well as the information content in sellers' marketing campaign, can then be aggregated into public *reputation measures*. In the spirit of Akerlof (1970), such measures might help welfare improving trades to occur, and in the process allow "good" sellers to increase their revenues. Second, precisely this public and non-anonymous nature of reputation can give rise to strategic behavior that affects observed market outcomes.

This paper tackles two relevant questions related to the incentives characteristic for on-line markets. We first investigate the role of *strategic competition* between sellers in altering the incentives to *build a reputation*, and how this affects observed market outcomes. To our knowledge, this is the first paper to characterize the interaction between strategic competition and reputation building. Second, we analyze quantitatively two identification problems in the reduced-form estimation of the impact of reputation on a seller's revenues. These problems can arise when strategic or reputation building behavior is not incorporated into the estimated model, even when panel data or field experiment data on revenues and reputation measures is used.

In our benchmark environment, two long lived sellers repeatedly auction homogeneous items to a finite number of different bidders each period. Over time, sellers compete in attracting a higher number of high valuation bidders. Sellers have different abilities, a feature that is only privately observed. In addition, once the good is sold, sellers put effort in delivering the item (i.e. send the promised good on time, without quality deficiencies, etc.), a decision unobserved by other buyers and sellers. The game therefore presents both adverse selection and moral hazard problems. Buyers have private valuations for the good on auction, and do not have any incentive to purchase more than one object over time. Once the transaction is completed, the winner of the auction *truthfully* reports the outcome in a binary form, an information which is added to the public history of the seller. Buyers form beliefs in a Bayesian fashion about sellers' unobserved abilities and anticipate the optimal effort decision for each type of seller. Based on these measures, they optimally select an auction and place a bid as described in Bapna et al. (2010).

Our first finding points out the fact that when sellers compete strategically, competition erodes marginal future profits and hampers sellers incentive to exert high effort in the first place: the higher the current reputation of the competitor, the less effort is a seller willing to put in order to build a reputation for the future. Nevertheless, we find that, as the number of bidders in the market increases, the effect of competition on effort exertion is reduced. In addition, a high number of potential buyers every period incentivizes sellers to work more in the present as they expect higher future profits (i.e. a higher highest expected bid in their auction). Our results on reputation building per se fall in line with Cripps et al. (2004) and Holmstrom (1999) predictions. Independent on whether a retailer acts as a single seller or faces competition from another fellow, a seller will always choose a relative higher effort to improve on his reputation when the latter is low. Once his type has been learned by the market, he opts for a relatively lower effort. This has been coined as the "spending of reputation".

Our second contribution consists in analyzing two identification problems that are linked to the reduced form estimation of the impact of reputation on revenues when strategic interactions and reputation building behavior are not taken into account in the estimation process. In addition, we analyze the dependence of such bias on market characteristics, such as the number of bidders at any given point.

Recent empirical studies have estimated different reduced form effects of reputation and feedback changes on sellers' revenues / sales in on-line markets, using both randomized "field" experiment data and observational panel data on revenues and reputation. Resnick et al. (2006) presents the first field experiment study in which a seller with an established reputation is asked to create a second identity and to operate (i.e. sell) under both identities employing the same "effort". The researchers then compare different revenue measures generated under the two identities.[1] This strategy allows them to control for many potential factors omitted in observational studies, related specially to the effort in presenting the good, in answering inquiries, in packaging, etc. They show that, indeed, the identity with better reputation generates higher revenues: willingness to pay increases by 8%, although they also find that one or two negative feedback does ot have a significant effect on revenues for the "new" seller. In observational data study, Cabral and Hortaçsu (2010) analyze a sample of 819 sellers of three different goods[2] and follow them at monthly intervals between October 24, 2002 and March 16, 2003. They first estimate a reduced form impact of reputation on prices using cross sectional data; in a second exercise, they use longitudinal data of seller's sales and reputation histories to estimate the impact of the *first negative feedback* on the dynamics of sales. They find that conditional mean (age-detrended) sales growth rates decrease from $+5\%$ to -8% immediately after the *first* negative.[3]

We claim, however, that such reduced form (linear) estimations of the impact of reputation miss two crucial aspects related to reputation building and strategic competition: both sellers' reputations and unobserved effort decisions might affect current revenues of a given seller, being at the same time correlated with the seller reputation. This feature is central to our model, as we consider it to have a significant impact on a seller's behavior. In order to understand the importance of such omission, we study the infinite horizon version of the game described above. This allows us to focus on the dynamics of revenues, efforts and reputations. We modify the computational algorithm presented in Doraszelski and Pakes (2007), and solve numerically

[1] They also document 15 previous observational studies of the impact of reputation on different measures of revenues.

[2] IBM ThinkPad T23, collectible coins and 1998 Holidays Teddy Beanie Babies.

[3] In order to estimate these effects, Cabral and Hortaçsu (2010) averaged weekly sales growth rates over a four week window before and after the week in which the seller got his first and respectively second, third, fourth and fifth negative feedback. Then they conduct t-tests for equality of means before and after the feedback. They use both a cross-section regression and a panel data approach to estimate the effect of reputation. The results, in line with previous research, show a steady positive trend of profits until the first negative feedback is received.

for a Markov Perfect Equilibrium. We then use simulated histories of revenues, reputation and effort to characterize the problems in identifying the reputation-elasticity of revenues from not incorporating reputational information and unobserved effort decisions into a reduced form model. In our dynamic game, the relevant elasticity is non-monotone on a seller's reputation, and it varies with the competitor's reputation. Due to the strategic behaviour, reputations are correlated in equilibrium, which implies that the omission of the competitors reputation in the estimated model will bias the results. The mechanism that leads to this bias arises if the decisions made by a seller's competitors depend on the seller's reputation, and if buyers somehow understand this.

The paper proceeds in the following way: in section 2 we briefly review our contribution from the point of view of recent papers on the topic. In section 3 we lay out the two-period environment and we define the equilibrium concept employed throughout the paper. section 4 then develops the main analysis, characterizing the mechanisms relevant for our results in a two period version of the game.[4] Finally, section 5 concludes.

2 Related Literature

The present paper makes a significant contribution to a recently emerged strand from the reputation literature, that analyses competition among long-lived firms in a dynamic environment dominated by adverse selection and moral hazard problems. Although the role of reputation building has been studied extensively since the seminal paper of Kreps and Wilson (1982), there is only a small number of studies analysing games with asymmetric information in which many long lived players compete against each other. This literature, fostered by Horner (2002), Tadelis (2002), Vial (2010) and Vial and Zurita (2012), builds on the one-seller case studied in the papers of Holmstrom (1999) and Mailath and Samuelson (2001). The purpose of these mod-

[4] In the appendix to this chapter, Appendix E which is preliminary work, I describe and briefly characterize an infinite horizon version of the game in order to assess quantitatively the the problems to empirically identify the reputation elasticity of revenues.

els is understanding whether competition can induce a reputational equilibrium in which opportunistic players choose high effort. The way in which these studies model competition, however, is through Walrasian markets. In consequence, the strategic behaviour of the long-run players becomes rather limited, which implies that sellers do not incorporate in their decisions the market's beliefs about their competitors. We contribute to this literature by assuming a dynamic auction environment with a finite number of sellers and buyers; this allows us to explicitly analyze the role of *strategic competition* in shaping sellers' reputation building decisions.

Cripps et al. (2004) and Holmstrom (1999) show that within a monopolistic environment a reputational mechanism cannot survive indefinitely as a mean to sustain an equilibrium with a high level of effort if the uncertainty about types disappears over time. Reputation - in the sense of Kreps and Wilson (1982)- is a commitment device that allows sellers to overcome the moral hazard problem, characterized by the short-term incentive to exert low effort, even though higher profits could be earned when committing to high effort in the long run. The dynamics of reputation in our benchmark game between a monopolist and a sequence of N-bidders sets is subject to the same pattern: as long as there is uncertainty about one's type, sellers exert costly effort in equilibrium. Over time, the market learns the seller's type and so the incentives to employ costly effort gradually decrease. As in Holmstrom (1999), the type of a seller has an intrinsic value, in addition to the information that it provides about the effort decision of a seller in equilibrium.

Our benchmark game stems from Mailath and Samuelson (2001), with some important distinctions. They consider an environment with imperfect public monitoring, in which consumers observe only a noisy signal of firms' decision. "Inept" sellers can only exert low effort while the "competent" have the choice between high or low effort. Reputation building then becomes an exercise where competent firms try to separate themselves from inept ones, rather than mimicking a "Stackelberg" type. The key ingredient for maintaining high effort on the equilibrium path is the replenishment of uncertainty by adding an unobservable type change process. With permanent types, they find that there are no pure strategy Markov equilibrium with high effort is exerted in equilibrium. The key idea behind these results lies in the convergence of high and low effort value functions when uncertainty is not replenished or when firms are replaced always by competent ones. Our benchmark game does not present such uncertainty in types, and we can still generate temporarily high effort. The reason lies in that, unlike Holmstrom (1999) or our paper, in Mailath and Samuelson (2001) a high reputation is valuable only to the extent that it signals high effort.

Horner (2002), Vial (2010) and Vial and Zurita (2012) analyze reputational equilibrium in competitive environments. In a setup in which both firms and consumers are atomistic and infinitely lived as described in Horner (2002), competition supports the existence of equilibria in which opportunistic firms always exert high effort. In Horner (2002) reputation is built gradually and vanishes instantly upon the deliverance of bad quality, for which a firm is punished by losing its customers to other firms with impeccable records. Prices are initially low but rise with a firm's reputation. Vial (2010) and Vial and Zurita (2012) consider a competitive environment as in Mailath and Samuelson (2001) and show that a reputational equilibrium with high effort can be attained. In contrast with Horner (2002), in Vial (2010) and Vial and Zurita (2012) reputation builds and dissipates gradually. Nevertheless, while Vial (2010) focuses on the population distribution of reputations and its impact on equilibrium prices, Vial and Zurita (2012) investigates the entry/exit decision of a firm and the resulting industry dynamics. As in Horner (2002), sellers in our model profit more not only from having *a good* reputation, but from having *the better* reputation. Under the assumption that the number of buyers is sufficiently higher relative to the number of sellers, we ensure that sellers will always expect a certain number of bidders with positive probability. The allocation of bidders to auctions in equilibrium is done in an assortative fashion: bidders with high valuation bid exclusively in the auction of the seller with the *relatively higher* chances to successfully deliver the good, while bidders with valuations from the lower part of the distribution bid with positive probability in both auctions. The threshold that separates the "high valuation" bidders from the "low valuation" bidders is endogenously determined by the likelihood ratio of successful deliveries of the two sellers. This result ensures a gradual increase and respectively a gradual dissipation of reputation as in Vial (2010). This very aspect will allow us to investigate the strategic role competition plays in our model in contrast with the above mentioned papers.

As in Vial (2010) and in our model, the equilibrium price function is key to understanding the incentives of sellers to exert effort, both in the monopolistic and in the competitive setting. The equilibrium price is a function of both equilibrium beliefs about a seller's type and the distribution of buyers' valuations and it incorporates the equilibrium behavior of both parties. In this setting, the updating of beliefs after a success or failure determines both the effect of current actions on future reputations and implicitly on future revenues. The main difference with respect to Vial (2010) and Horner (2002) is our focus on the *finite* number of bidders and sellers, that induces long-run players to strategically compete for high valuation buyers.

3 The Set-Up

The general game builds on Mailath and Samuelson (2001) and introduces some modifications. In the baseline game, time is discrete, and indexed by $t = 1, 2$. In each period, there are either 1 or 2 long-lived seller(s) that open an auction each, and N new risk-neutral buyers that select (if more than 1 seller) and bid once in the auctions available at the time they arrive to the market. A good is allocated to a buyer through a second-price auction. All auctions start and end simultaneously. Further, the set of buyers will be denoted by $\mathcal{N} = \{1, 2, \ldots, N\}$, while the set of sellers by $\mathcal{J} = \{1, 2\}$.

3.1 Sellers

There are 2 risk-neutral long-run sellers that each period put up for sale a homogeneous good. Sellers are long-lived, with a *common* discount factor δ. A seller can be of either of"good" or of "bad" type. The type is indexed by $\theta \in \{g, b\}$, where

$0 \leq b \leq g \leq 1$ and stands for the *privately* observed innate ability or trustworthiness of the seller to execute a successful transaction. The *prior* probability that a seller is of good type is $\mu_{j,0} \equiv P(\theta_j = g)$.

Assumption 1 *The type θ_j is fixed throughout the life of the seller j, for $j = 1, 2$.*

Assumption 1 implies that there is no "replenishment" of uncertainty. This is different from Mailath and Samuelson (2001), where types are allowed to change randomly or endogneously.

In our setting, we think about the type of a seller as an "aggregate" measure of *competence* and *morality* characteristics.[5]

3.1.1 Actions, Information and Payoffs

For successfully delivering the product after the auction has been carried out, a seller has to exert some effort e. We think about the effort as the amount of work put in delivering the product properly, securely and professionally packaged, in sending in the specified time the exact advertised object. Effort has a cost for the seller, $c(\cdot)$, and is *unobserved* by buyers and by the other seller (i.e. the competitor).

Assumption 2 *The cost function is increasing and convex and has the following properties: $c(e^\theta) = 0$, $c'(e^\theta) = 0$ $\forall e^\theta \in [0, \underline{e}]$ and $c(e^\theta) > 0$, $c'(e^\theta) > 0$ and $c''(e^\theta) > 0$ $\forall e^\theta \in (\underline{e}, 1)$. The cost function is the* same *for all sellers.*

In other words, for a sequence of low efforts, the seller incurs no costs; effort becomes costly above the threshold $\underline{e}$, which we call the "maximum costless effort".

[5]When interpreting a seller's type, we don't distinguish among two *dimensions* of her privately observed characteristics: her *trustworthiness* in revealing information before trade takes place and her *ability* to deliver the good in a successful way. Jullien and Park (2011) analyzes the emergence of reputational equilibrium when these two dimensions are distinguished. They propose an equilibrium where there is an endogenous complementarity, in terms of market perception of information, between honesty in pre-trade communication (honesty) and ability to deliver good quality (reputation). In such setting, pre-trade communication is *cheap-talk*; however, in the benchmark equilibrium they consider, high type sellers truthfully reveal the quality of their good, while low types tend to *lie* more often; once the quality of the good is revealed (through a feedback system) a truthfull pre-trade communication is mapped into a higher probability of the seller being of high ability. Therefore, learning occurs at a faster rate than when the distinction in the dimensions of reputation is not consider (our paper).

By devising a cost function of this form, we are implicitly assuming that it is a weakly dominant strategy for each seller to exert a positive amount of effort, to which a strictly positive probability of success is associated [6].

Each transaction carried out by seller j and buyer i at time t has an *outcome*, which we define as a random variable $y_{t,j}$ that can take two values: S (success) or F (failure).

Assumption 3 *The technology is such that the probability of success depends both on sellers' effort* and *on his type, i.e. the probability of success of each transaction takes the form*

$$P\left(y_{j,t} = S|e_{t,j}^{\theta_j}, \theta_j\right) = e_{t,j}^{\theta_j} \cdot \theta_j \tag{1}$$

For a given effort level, the "good" seller has almost always a higher probability of success in delivering the product. Conditional on effort and types, outcomes are independent across sellers:

$$P\left(Y_j = y_j, Y_{-j} = y_{-j}\middle|\theta_j, \theta_{-j}; e_j^{\theta_j}, e_{-j}^{\theta_{-j}}\right) = P\left(Y_j = y_j\middle|\theta_j, e_j^{\theta_j}\right) P\left(Y_{-j} = y_{-j}\middle|\theta_{-j}, e_{-j}^{\theta_{-j}}\right) \quad \forall j \neq -j \tag{2}$$

At the beginning of each period t, sellers have two sets of information: *public* information, given by the publicly observable sequence of outcomes $\{y^{t-1}\}_{t\geq 1}$ with $y^{t-1} = \{y_{1,t-1}, y_{2,t-1}\}$ assuming that $y_{j,0} = \emptyset \ \forall j \in 1, 2$, and *private* information, given by his type (which is time-invariant) and the sequence of own effort decisions $\left\{e_{j,t}^{\theta_j}, \theta_j\right\}_{j,t\geq 1}$. [7]

The public *reputation* of seller j at the beginning of auction $t \geq 1$ is defined as

$$\mu_{j,t-1} \equiv P\left(\theta_j = g|y^{t-1}\right)$$

[6] Note that this is a similar assumption to a standard one in the literature, which allows for 0-effort but assumes that the probability of success (or more generally, the density of outcomes) is always greater than 0. In particular, many papers assume two possible effort levels, say e^L and e^H, and then assume that the density of outcomes given e^H first-order stochastically dominates the one given e^L.

[7] Here we assume that prices paid by previous buyers in seller's j past auctions are not observed by upcoming buyers. This rules out complicated behavior and equilibria in which buyers condition their bidding on past realized prices.

where $\mu_{j,0}$ is the belief about the seller's j type at the beginning of the game. Given our assumption regarding the public information y^{t-1} about seller j at the beginning of period t, $\mu_{j,t-1}$ is the *common* belief that all other players (buyers and other sellers) hold about seller j being of good type.

Let $\mathcal{H}_j^t$ be the set of all possible public histories seller j can have at the beginning of period $t+1$: $\mathcal{H}_j^t = \{h_j^t \in \{S, F\}^t, t \geq 1\}$, where $h_j^0 = \emptyset$. The set of all possible histories of the game is defined by the product of all sellers' histories $\mathcal{H}_t = \underset{j}{\times} \mathcal{H}_j^t$.

Seller's j expected revenue from an auction, R_j, is the expected second highest bid of the respective auction paid by the winning bidder. In a second price auction, the expected revenue of the seller is given by

$$\boldsymbol{R}_j = \mathbb{E}\left(b_j^{(2)} \middle| \, \# \text{ of buyers, strategies of buyers}\right)$$

Henceforth, the expected revenue of seller j at time t denoted by $R_{j,t}$ is a function of all sellers' equilibrium actions in period t. Under the assumption that each seller sells only one good each period, seller's j ex-ante stage payoffs are given by

$$\pi_{j,t}^{\theta_j} = \boldsymbol{R}_{j,t} - c\left(e_{j,t}^{\theta_j}\right) \tag{3}$$

The inter-temporal problem for seller j of type θ_j is then

$$\max_{\left\{e_{j,t}^{\theta_j}\right\}_{t=1,2}} \Pi_{j,0}^{\theta_j} = \boldsymbol{R}_{j,1} - c\left(e_{j,1}^{\theta_j}\right) + \delta \mathbb{E}_1\Big(\boldsymbol{R}_{j,2} - c\left(e_{j,2}^{\theta_j}\right)\Big) \tag{4}$$

3.2 Buyers

The benchmark setting follows closely Bapna et al. (2010). Different from them, however, in our setup buyers anticipate seller's equilibrium actions and incorporate this information in their optimal bidding strategies.

Buyers have *unit demand* and only live for one period, i.e. they are interested in buying one unit of the good offered by sellers today (in later sections we will discuss deviations from this assumption). Each buyer has private information about his valuation v^i, and valuations are independently drawn from the distribution $v^i \sim U[0,1]$.

Assumption 4 *The total number of bidders N is constant in every period. In addition, both N as well as the distribution from which valuations are drawn every period are* common *knowledge to all market participants.*

3.2.1 Actions and Payoffs

Buyers' strategies consist in deciding in which auction to bid, and the bid $b^i_j \in [0, \infty)$ they place in the respective auction. Therefore, a buyer's i strategy consists of two elements: a *bid vector* $b^i(v^i)$ and an *auction selection strategy* $s^i(v^i)$. The *bid vector* $b^i = (b^i_1, \ldots, b^i_J)$ consists of elements b^i_j representing the amount of money buyer i is willing to place in seller's j auction if she were to bid in that auction. An *auction selection strategy* is a vector $s^i(v^i) = (s^i_j(v^i))^J_{j=1}$ where $s^i_j(v^i)$ is the probability that buyer i with valuation v^i bids in seller's j auction.

Assumption 5 *In our benchmark game, buyers are allowed to select only* ***one auction*** *and to bid once.*

Since buyers' valuations are freshly drawn every period, their strategic behavior is independent of time and hence we omit the time index in the buyers' problem.

Prior to selecting a seller, each buyer forms beliefs about sellers delivering successfully the good. We make a final assumption about the information available to buyers:

Assumption 6 *Current bidders cannot observe past prices.*

Assumption 6 implies that buyers only observe past outcomes before deciding. Given sellers' reputations $\mu = (\mu_j)_{j \in \mathcal{J}}$, buyers anticipate the effort a seller j of type θ_j will exert. The belief regarding the successful delivery of seller j is defined as

$$r_{j,t} \equiv P\left(y_{j,t} = S | \mu_{j,t-1}\right) = \mu_{j,t-1} \cdot g\tilde{e}^g_{j,t} + (1 - \mu_{j,t-1}) \cdot b\tilde{e}^b_{j,t} \tag{5}$$

where $\tilde{e}^{\theta_j}_{j,t}$ is the anticipated effort of seller j in auction t if she were of type θ_j. In other words, $r_{j,t}$ stands for the anticipated probability of a successful transaction when facing a seller of unknown type whose public beliefs are by $\mu_{j,t-1}$.

It is a well established result that in a second price auction truthful bidding is a weakly dominant strategy. In our setup buyers' optimal strategy is to bid their expected valuation of the item in each auction:

$$b_j^{*i}(v^i, r_{j,t}) = r_{j,t}v^i \tag{6}$$

Conditional on winning auction j, buyer's i expected surplus is:

$$\tilde{U}_i\left(v^i, r_{j,t}, b_j^{-i}\right) = r_{j,t}v^i - \mathbb{E}\left(b_j^{(2)}|b_j^i > \max_{-i}\left\{b_j^{-i}\right\}\right) \tag{7}$$

where $b_j^{(2)}$ is the second highest bid in j's auction and $\{b_j^{-i}\}$ is the set of all bids in seller's j auction j, excluding bidder i.

Equation (7) takes into account only the subset of buyers that have decided to bid in seller's j auction. The ex-ante expected surplus from bidding in seller's j auction is given by

$$U_i\left(v^i, r_{j,t}\right) = \sum_{n=0}^{N-1} P\left(n+1 \text{ bidders in auction } j\right) P\left(v^i = \max_k v^k\right) \times \left(r_{j,t}v^i - \mathbb{E}\left(b^{(2)}|b^{(1)} = b^i, n+1 \text{ bidders}\right)\right) \tag{8}$$

The first term in expression 8 is the probability that given that there are $n+1$ bidders in seller's j auction, bidder i has the highest valuation among all the bidders k in that auction. The second term stands for the expected instantaneous expected surplus of bidder i given that he wins the auction of seller j in which there are in total $n+1$ bidders.

Buyer's i expected surplus from bidding in seller's j auction increases both in seller's j reputation and in the expected distance between the highest and the second highest valuations in seller's j auction. Hence, buyers face a tradeoff: on the one hand, a buyer has a higher expected revenue from bidding in the auction of a seller with a relatively high reputation, since this implies a higher probability of receiving the good, conditional on winning the auction. On the other hand, given his valuation v^i, bidding in the auction of lower reputation sellers might bring a higher surplus and a higher probability of winning if buyers of relative low valuation bid in these auctions.

3.2.2 Beliefs Updating

Buyers are forming and updating their beliefs about the type of seller they are facing in a Bayesian fashion. Having observed the history of seller j consisting in the sequence of outcomes of past transactions up to time t, buyers hold *common* beliefs, $\mu_{j,t-1}$, at the beginning of the t-th period about seller j being of "good" type. After the realization of the outcome of period t, sellers and buyers of period $t+1$ revise their beliefs about other sellers' types using the Bayes rule. In case of a success, the beliefs at the beginning of period $t+1$ are update to $\mu_{j,t} = \mu_{j,t-1}^{S}$, while in the case of a failure the public believes that seller j is good with probability $\mu_{j,t} = \mu_{j,t-1}^{F}$. Concretely,

$$P_t\left(\theta_j = g \middle| y_{j,t} = S, \mu_{j,t-1}; \left\{\tilde{e}_{j,t}^{\theta_j}\right\}\right) \equiv \mu_{j,t-1}^{S} = \frac{\mu_{j,t-1} g \tilde{e}_{t,j}^{g}}{\mu_{j,t-1} g \tilde{e}_{j,t}^{g} + \left(1-\mu_{j,t-1}\right) b \tilde{e}_{j,t}^{b}} \quad (9)$$

and

$$P_t\left(\theta_j = g \middle| y_{j,t} = F, \mu_{j,t-1}; \left\{\tilde{e}_{j,t}^{\theta_j}\right\}\right) \equiv \mu_{j,t-1}^{F} = \frac{\mu_{j,t-1}\left(1-g\tilde{e}_{j,t}^{g}\right)}{\mu_{j,t-1}\left(1-g\tilde{e}_{j,t}^{g}\right) + \left(1-\mu_{j,t-1}\right)\left(1-b\tilde{e}_{j,t}^{b}\right)} \quad (10)$$

3.3 Markov Strategies

As specified above, the public history of a seller records only the outcomes of past transactions. Such information restriction when describing strategies still entails many equilibrium in which reputation has different roles. In order to rule out non intuitive equilibria, we will focus on Markov strategies, i.e. we want strategies to make behavior by each seller at any t only dependent on a relatively small subset of the public information available, or on a sufficient statistic of it.[8]

[8] Given the functional form for the conditional probability of success, $P\left(y = S|\theta, e\right) = \theta e$, then for any equilibrium, every possible history of delivery outcomes $h^{t-1} \in \mathcal{H}^{t-1}$ will have positive probability in such equilibrium. This will simplify considerably the equilibrium characterization, since there will be no "off-equilibrium path" information set.

Assumption 6 implies that, although sellers can observe past prices, they will turn out not to be payoff relevant variables at the time of deciding about effort.

In the case of on-line auction platforms such as eBay, the information available to sellers and potential buyers about each other is significantly limited. Usually, such information amounts to *feedback* scores (current and historical, up to a certain past date) which represent a summary statistic of past evaluations by buyers or sellers. This information is public, and is the *same* information observed by a researcher / economist / econometrician.

Given assumptions 1-6 and the above considerations, the reputation of seller j at the beginning of period (auction) t, $\mu_{j,t-1}$, is a *sufficient statistic* about that seller's type: it summarizes all the payoff-relevant information about seller's j type available in the only publicly observed variables y_j^t. Such information is payoff-relevant due to the fact that it helps predict the behavior of sellers in the current auction (recall that buyers are short lived in our benchmark game). Note that, given the finite-time nature of the game, then the state space will also include the time period t.

Given the above, a Markov strategy for seller j is :

$$e_{j,t} : [0,1]^2 \times \Theta \times T \longrightarrow [0,1]$$

where $e_{j,t}(\mu_1, \mu_2; \theta_j) \equiv e_{j,t}^{\theta_j}(\mu_1, \mu_2)$ is the effort choice of a seller with type θ_j at period t, when buyer's beliefs about all sellers' types are described by the vector $\{\mu_j\}_{j=1}^2 \in [0,1]^2$.

When facing sellers with different reputations, buyers' strategies consist in an *auction selection vector* and in a *set of bids*. Given that we restrict buyers to bid in only one auction, an auction selection strategy for buyer i is a mapping

$$s^i : [0,1]^2 \times [0,1] \longrightarrow [0,1]^2$$

where $s^i(v^i; \mu_1, \mu_2) \equiv \left\{ s_j^i\left(v^i; \mu_1, \mu_2\right) \right\}_{j=1}^2$ is a 2-dimensional vector of selection probabilities.

3.4 Timing

The timing is presented in figure 1. The game proceeds in the following way: at time $t = 0$, the game is in state $(\mu_{j,0})_{j\in\mathcal{J}}$, where $\mu_{j,0}$ $j \in \mathcal{J}$ are the publicly held time-0-beliefs (prior beliefs) about sellers being competent. We allow these beliefs to be different across sellers. Sellers open simultaneous auctions in which all sell the same homogeneous good. Buyers simultaneously select an auction and place a bid. The winners of each auction are announced and the second highest bid in each auction is paid out to the respective seller, as described in the previous section. Sellers decide then on the effort they want to put in delivering the object, bearing in mind that the probability of a success is proportional to their effort and that the success cannot be guaranteed. Any of the two possible outcomes - success $y_{j,0} = S$ or failure $y_{j,0} = F$ - will be then truthfully reported by buyers and enter the public record. Based on the outcome, other sellers and buyers update their beliefs about seller's j type. The infinite game consists of infinitely many repetitions of the stage-game described above.

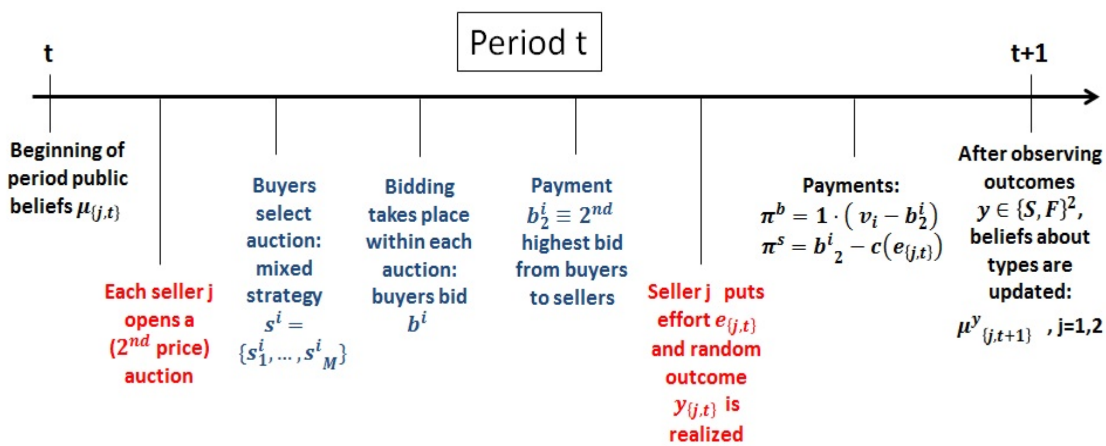

Figure 1: Timing

3.5 Equilibrium

Buyers' equilibrium behavior derived in Bapna et al. (2010) is based on their perceived success probability $r_{j,t}$. In our setup, buyers observe sellers' reputation $\mu_{j,t-1}$ and, anticipating sellers' equilibrium behavior $\tilde{e}^{\theta_j}_{j,t}$, buyers form a measure of the success probability $r_{j,t}$for each seller, as described in equation 5. Note that the buyer's problem and equilibrium behavior can be characterized for a *given set of beliefs*. In other words, buyers play a one shot game and their behavior is static by assumption.[9]

Definition 7 *A tuple* $\left(\left\{e_j^g(\boldsymbol{\mu}), e_j^b(\boldsymbol{\mu})\right\}_j, \left\{s^i(v^i;\boldsymbol{\mu}), b^i(v^i;\boldsymbol{\mu})\right\}_i, \boldsymbol{\mu}\right)$, *with* $\boldsymbol{\mu} \equiv \{\mu_1, ...\mu_J\}$, *is a* ***Markov Perfect (Bayesian) Equilibrium*** *(MPE) of the reputation game with asymmetric information if, for a buyer's valuation distribution* $F(\cdot)$*:*

1. *Buyers selection strategies* $s^i(v^i;\boldsymbol{\mu})$ *and bidding behavior* $b^i(v^i;\boldsymbol{\mu})$ *are optimal given* $\boldsymbol{\mu}$*, their valuation* v^i *and distribution* F*.*

2. *Sellers choose effort* $\left\{e_j^g(\boldsymbol{\mu}), e_j^b(\boldsymbol{\mu})\right\}_j$ *to maximize* ***continuation*** *payoffs* (4) *, given* $\boldsymbol{\mu}$*.*

3. *Sellers' behavior is consistent with the* ***Markov-belief function*** (5)*:*

$$r_{j,t} = \mu_{j,t-1} \cdot g e_{j,t}^g + (1 - \mu_{j,t-1}) \cdot b e_{j,t}^b$$

4. *Beliefs are updated according to Bayes' rule:*

$$\mu_{j,t}(\theta_j = g|, y_t, \mu_{j,t-1}) = \frac{P(y_{j,t}|\theta_j = g) \cdot \mu_{t-1}}{\sum_{\theta_j} P(y_{j,t}|\theta_j) \cdot P(\theta_j)} \quad \forall y_{j,t} \; if \; \sum_{\theta_j} P(y_{j,t}|\theta) \cdot P(\theta_j) > 0.$$

4 Characterization and Analysis

In order to disentangle the effects that competition and reputation incentives have on effort an welfare, we study separately two versions of the game: (i) the case when there only one seller (and everyone knows so) , and (ii) the case with 2 sellers.

4.1 Reputation Effects with One Seller

We first analyze the two period game with a *single* seller. This will allow us to isolate how the two main information asymmetries shape the equilibrium effort strategies and reputation building. We do this by characterizing the market equilibrium in such

[9]We analyze the role of assuming short lived buyers in the robustness section.

scenario and then comparing with three alternative informational assumptions: (i) a situation in which the seller can *commit ex-ante* (i.e. before buyers bid) to exert some effort, and where the type (ability) of the seller is observed; (ii) a second scenario where the seller can commit, but her type is now unobserved; and (iii) our case of interest, in which the seller cannot commit ex-ante, and her type is unobserved, i.e., where both adverse selection and moral hazard problems are present.

Our results are consistent with a (finite horizon version) of Holmstrom (1999) and Mailath and Samuelson (2001): there is an equilibrium (which in our case is unique in Markov strategies) in which the monopolist *temporally* exerts costly effort, and where such effort strategy is non-monotone in initial beliefs (the state variable).

Building a Reputation in The First Period

A seller auctions off a homogeneous good over two periods. Given the timing in the game, the seller will employ the maximal costless effort $\underline{e}$ in period 2 as there are no future payoffs from which he can gain when employing a costly effort, independently of her type and on the outcome of the first period.

Since there are no alternative auctions, buyer's selection strategy is trivial. Equilibrium bidding implies that the expected revenue of the seller in a second price auction in which valuations are drawn from an uniform distribution and the total number of buyers is N equals:

$$E\left(b_t^{(2)}\right) = r_t \times H_2\left(N\right) \tag{11}$$

where $E\left(b_t^{(2)}\right)$ is the expected second highest *bid*, and $H_2\left(N\right) \equiv \mathbb{E}\left(v^{(2)}|N \text{ bidders}\right)$ is the expected *valuation* of the second highest bidder (a second order statistic; see appendix B for the derivation of the functional form for $H_2\left(N\right)$).

There are two important features of $H_2\left(N\right)$: (i) it is a function of N, the number of buyers each period; and (ii) it does not depend on the seller's reputation, a feature that will not be necessarily true once we add competition between sellers, and which will drive seller's strategic behavior. From now on, we omit the explicit dependence on N. Equation (11) therefore implies that the expected second highest bid will be a linear function in monopolist's reputation.

Upon receiving the money, seller 1 decides how much effort to put in delivering the item. The instantaneous expected payoff of the monopolist at the beginning of period t, conditional on her type θ is given by:

$$\pi_t(\theta, e_t^\theta, \tilde{e}_t^\theta, \mu_t) = E\left(b_t^{(2)} \mid \mu_t\right) - c(e_t^\theta) \tag{12}$$

The present value (PV) of a seller's profits conditioning on her type θ is then

$$\Pi_0^{monop}\left(\theta, e_1^\theta, \tilde{e}_1^\theta, \mu_0\right) = \pi_1(\theta, e_1^\theta, \tilde{e}_1^\theta, \mu_0) + \delta\mathbb{E}\left[\pi_2\left(\theta, \underline{\mathrm{e}}, \mu_1\right)\middle|\theta, e_1^\theta\right] \tag{13}$$

where

$$\mathbb{E}\left[\pi_2\left(\cdot\right)\middle|\theta, e_1^\theta\right] = e_1^\theta\theta\cdot\pi_2(\theta, e_2^\theta, \tilde{e}_2^\theta, \mu_1|y_0 = S) + \left(1 - e_1^\theta\theta\right)\cdot\pi_2(\theta, e_2^\theta, \tilde{e}_2^\theta, \mu_1|y_0 = F) \tag{14}$$

Equilibrium effort levels for each type of seller have to solve

$$\max_{e_1^\theta} \; \Pi_0^{monop}\left(\theta, e_1^\theta, \tilde{e}_1^\theta, \mu_0\right) \tag{15}$$
$$s.t. \;\; e_t^\theta \in [\underline{\mathrm{e}}, 1]$$

Best responses for each type of seller are characterized by the system of first order conditions for $\theta = \{g, b\}$:

$$c'(e_1^{*\theta}) = \delta\theta\left(\pi_2^S - \pi_2^F\right) = \delta\theta H_2\left(N\right)(\mu_1^S - \mu_1^F)(g-b)\underline{\mathrm{e}} \tag{16}$$

where, as before, $H_2\left(N\right)$ is the expected second highest value in the seller's auction and $e_t^{*\theta}$ is the best reply of seller-type θ to the effort of seller-type $-\theta$. These are, in other words, optimal effort levels employed in period $t = 1$).

Lemma 1 *The single crossing condition yielded by the complementarity of effort and types, implies that higher types always choose higher efforts:* $e_1^{*g} > e_1^{*b}$.

Proof. Note that

$$c'(e_1^{*g}) = \delta g H_2(\mu_1^S - \mu_1^F)(g-b)\underline{e} > \delta b H_2(\mu_1^S - \mu_1^F)(g-b)\underline{e} = c'(e_1^{*b})$$

The cost function being monotonically increasing and convex results in $c'(e_1^{*g}) > c'(e_1^{*b})$, from which follows that $e_1^{*g} > e_2^{*b}$

With unobservable types, the *reputation* attributed to the seller at time $t = 2$ crucially depends on the outcome of the first period: $r_2 = (\mu_1 g + (1-\mu_1)b)\underline{e}$, where μ_1 are updated beliefs given the outcome of the first period ($\mu_1 = \mu_1^S$ or $\mu_1 = \mu_1^F$). The buyers active at this stage will bid their expected valuation for the good: $b^{*i}(v^i) = r_2 v^i$. Given that reputation r_2 is increasing in beliefs $\frac{\partial r_2}{\partial \mu_1} = (g-b)\underline{e} > 0$, a monopolist has high incentives to conclude period 1 with a success as $\mu_1^S > \mu_1^F$. Therefore, employing a higher but costly effort level in period 1 will raise the probability of a success and the implied future payoffs. In determining the optimal effort level, the monopolist will trade off the future benefits of a higher reputation with the present cost of effort.

Proposition 1 below establishes the existence and uniqueness of an equilibrium in the 2 periods game, in which the monopolist seller temporally exerts positive effort. In addition, proposition 2 characterizes the role of a reputation mechanism in such equilibrium.

Proposition 1 *An equilibrium of the two-periods game where a monopolist of unobservable type chooses (temporally) costly effort exists and is unique.*

Proof. See appendix D

Proposition 2 below characterizes the equilibrium with moral hazard and adverse selection

Proposition 2 *For any type θ, $\theta \in \{b, g\}$, in the Markov equilibrium with hidden effort and unobserved type:*

1. *For any $\mu \in [0,1]$ the equilibrium effort level in period 1 increases with the discount factor δ; i.e., $\frac{\partial e^\theta_{j,1}(\mu)}{\partial \delta} > 0$, $\forall j, \theta$.*

2. *For any $\mu \in [0,1]$ the effort level of the good type in period 1 is increasing in g and decreasing in b; i.e., $\frac{\partial e^{\theta}_{j,1}(\mu)}{\partial g} > 0$ and $\frac{\partial e^{\theta}_{j,1}(\mu)}{\partial b} < 0$. The opposite is true for the effort level of the bad type.*

3. *The equilibrium effort level in period 1 is non-monotone and concave in initial beliefs μ_0: $\frac{\partial e^{\theta}_{j,1}(\mu)}{\partial \mu_0} > 0$ for $\mu_0 < \mu^*$ and $\frac{\partial e^{\theta}_{j,1}(\mu)}{\partial \mu_0} < 0$ for $\mu_0 > \mu^*$. The threshold μ^* is a function of the different parameters of the game, and in particular, of g and b.*

Proof. See appendix D.

Result 1 in proposition 2 is straightforward. The higher the discount factor for a seller, the more relevant future revenues become for her, and therefore reputation building has a higher value. Result 2. captures an important aspect of on-line markets (and markets in general with imperfect information in a dynamic setting). Figure 2 presents the argument graphically: the plot shows the updating of type-beliefs, for an initial prior, after a successful transaction and after a failed one (equations (9) and (9)) . The left-hand side plot presents the case where $g = .9$ and $b = .1$, while the right-hand side plot presents the case of $g = .7$ and $b = .6.$: if the difference $g - b$ is close to 1, we are in a situation in which a seller can be either very able / willing, or almost not able at all, to deliver successfully. For any initial belief, a seller of good type has then a strong incentive to separate herself from a seller of bad type through relatively high levels of effort, the reason being that a successful outcome will reveal a lot of information about the sellers ability; clearly, this is true as long as this marginal benefit is higher than the marginal cost of effort. For a bad type seller (i.e. the incompetent seller) things are different. Although higher g creates incentives for her to try to mimic a good seller, the same happens if the value of b increases. The reason is that when $g - b$ is small and effort is not observable, it is harder for buyers to distinguish among sellers of different abilities. This implies that effort will pay relatively more to bad type sellers. The convexity of the cost function then drives

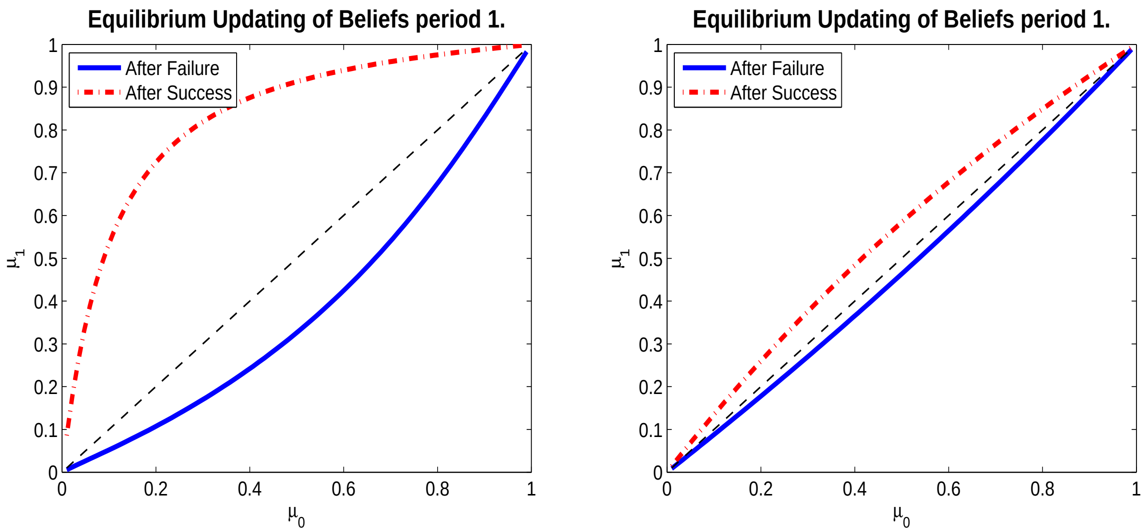

Figure 2: Equilibrium updating of beliefs after S and F. Left: $g = 0.9$ $b = 0.1$ Right: $g = 0.7$ $b = 0.5$

the low level of efforts in equilibrium. This is the standard likelihood ratio property coming into play.[10]

Result 3. requires a careful explanation. The upper row in figure 5 depicts how the optimal effort for both types varies with different initial beliefs. As long as the seller cannot commit ex-ante to an effort level, the uncertainty about the seller's type is what drives her incentives of employing costly effort. As the buyers believe with more certainty that they are facing a particular type the game starts resembling the one with observable types described above. It follows that the incentives for any seller type to exert costly effort vanishes. This is similar to the findings in Holmstrom (1999). In our game, this mechanism is clear from the optimality condition (16) for each type. As $\mu_0 \to 1$ also $\mu_0^S \to 1$ and $\mu_0^F \to 1$ so $\mu_0^S - \mu_0^F \to 0$. The equilibrium effort level for each type will converge then to the maximal costless effort level $e^{*\theta} \to \underline{\mathrm{e}}$. The same principle applies for the case where $\mu_0 \to 0$. The non-monotonicity in μ is close to the concept of "spending" a reputation described in Mailath and Samuelson (2001).

[10]See appendix C for a more detailed description of this.

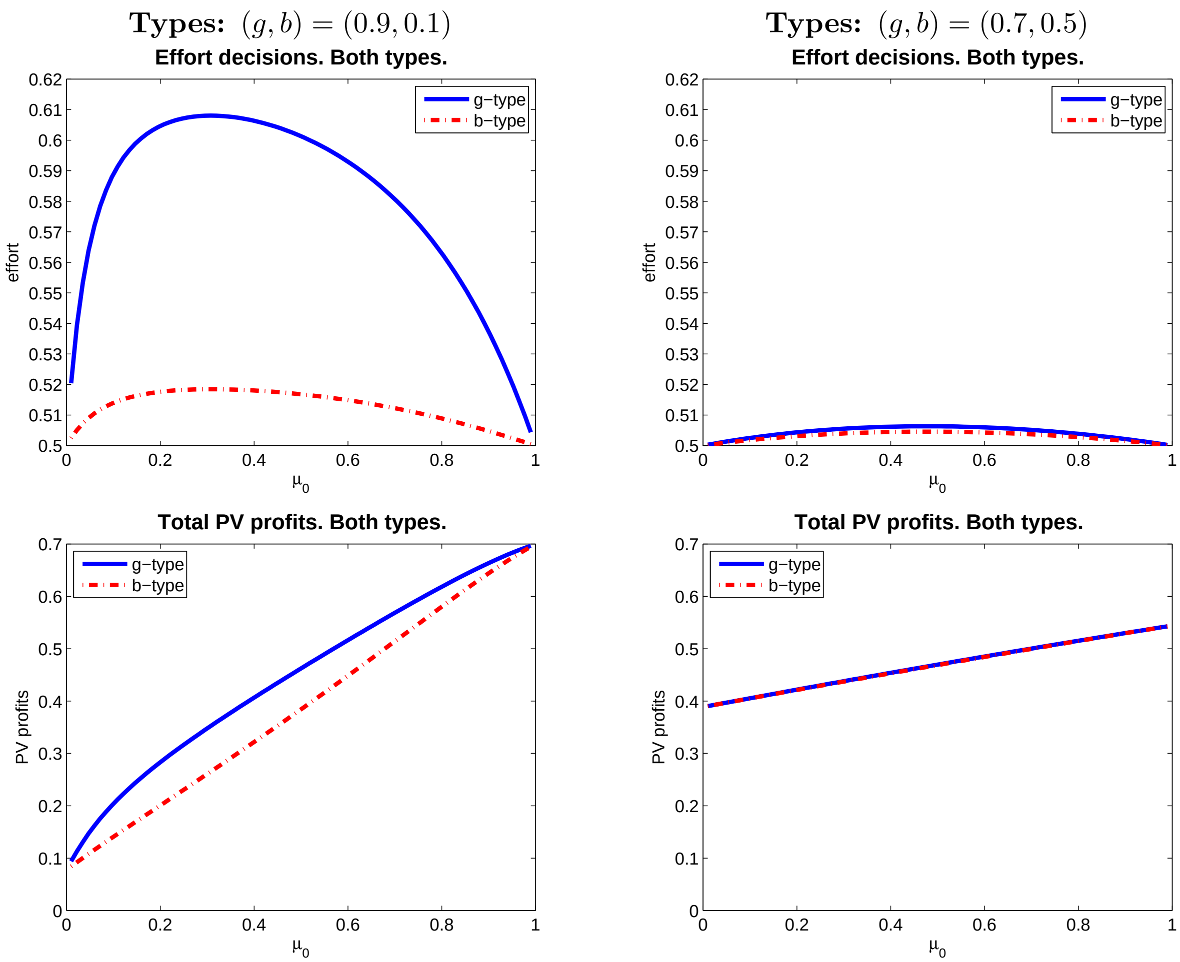

Figure 3: Equilibrium effort (top) and PV profits (bottom). Left: $g = 0.9$ $b = 0.1$ Right: $g = 0.7$ $b = 0.6$

4.1.1 Alternative Information Assumptions

If the seller's type is observable, the game is trivial. The only possible equilibrium is one in which the monopolist exerts the maximum costless effort in both periods, i.e. $e_t^{\theta obsNC} = \underline{\mathrm{e}}$, $t = 1, 2$, $\theta \in \{b, g\}$. Expected profits for the seller are then given by

$$\Pi^{\theta obsNC} = (1 + \delta)\left(\underline{\mathrm{e}}\theta H_2\right) \tag{17}$$

This result is straightforward: if the ability of the seller is publicly observable, the seller has no "reputation" to defend and maintain. The perceived probability of success of a transaction at time t is given by $r_t = \theta\underline{\mathrm{e}}$,$t = 1, 2$, $\theta \in \{b, g\}$. The corresponding bid in the first period by buyer i will be $b(v^i) = \theta\underline{\mathrm{e}}v^i$. Furthermore, the outcome of the first period is irrelevant for the second period since beliefs are degenerate. Employing costly effort has then no effect on future payoffs, and although they do affect the equilibrium bids in the first period (since buyers predict the seller's behavior), the seller is not able to commit to a given effort ex-ante. This implies an equilibrium effort that satisfies $c(e_t^{\theta obsNC}) = 0$. Independently of what had happened in period 1, the seller will always employ the maximal costless effort in period 2. Given the monopolist's type and anticipating her defective behavior, a buyer with valuation v^i will submit the following bid in the second period $b(v^i) = \theta\underline{\mathrm{e}}v^i$.

4.1.2 Efficient Effort

The efficient effort can be defined, given the buyer that has won the auction (the buyer with highest valuation $v^{(1)}$) , as a solution to the maximization of the total (ex-ante) surplus $\mathcal{W}_t$ of the transaction[11]:

$$\max_{\left\{e_1^\theta\right\}_\theta} \mathcal{W}_1 = -c\left(e_1^\theta\right) + \theta_1 e_1^\theta \mathbb{E}\left(v^{(1)}\right) \tag{18}$$

s.t.

$$e_1^\theta \in [0, 1]$$

[11] Recall that when there is just one seller all buyers bid in her auction; therefore, selection plays no role, which implies that we donot care about efficient allocation of buyers to sellers.

where $\mathbb{E}\left(v^{(1)}\right)$ is the expected highest valuation. The first order conditions of such problem for $\theta = b, g$ are

$$c'\left(e_1^{\theta eff}\right) = \theta \mathbb{E}\left(v^{(1)}\right) \tag{19}$$

In other words, for a given buyer's valuation $v^{(1)}$, the efficient allocation is such that a seller of good type should exert higher effort than a seller of bad type. In addition, (19) implies that, given the properties of $c(\cdot)$ and that $v_i \sim [0,1]$, $e_1^{\theta} > \underline{e}$. In other words, the efficient effort is greater than the maximal costless effort. The idea is simple: in our model, the seller does not value the good, while the buyer's value of the good is v. Given that the price payed is just a transfer between the buyer and the seller, efficiency stipulates that effort needs to compensate the marginal cost with a positive impact on the delivery technology.

Commitment and Observable Type

Consider first the case where a seller can credibly commit to a certain level of effort *before* the buyers bidding decision, and where her type is publicly observed / known. In such scenario, beliefs would play no role, and the probability of a success becomes $r_t = \theta e_t$. This implies that the problem for a seller is now a static one. Concretely, she would solve

$$\max_{e \in [0,1]} \theta e H_2 - c(e)$$

The FOC for such problem is

$$c'\left(e^{\theta obsC}\right) = \theta H_2 \tag{20}$$

The ability to commit allows the seller to shift her focus towards the *current* period: she now cares about the effect of her effort on the strategies of current buyers, and therefore drives the effort away from the maximal costless effort. However, note that $H_2 < \mathbb{E}\left(v^{(1)}\right)$: since the allocation mechanism is a 2nd price auction, the expected revenue for the seller incorporates the expected 2nd highest bid, which implies that $e^{\theta obsC} < e^{\theta eff}$, i.e. the effort decision under commitment and observable type would be lower than the efficient one.

What about seller's profits? For a seller of type θ, these are given by

$$\Pi^{\theta obsC} = (1+\delta)\left(e^{\theta obsC}\theta H_2 - c\left(e^{\theta obsC}\right)\right) \tag{21}$$

Lemma 2 *For any $\theta \in (0,1)$, $\Pi^{\theta obsNC} < \Pi^{\theta obsC}$. In addition, $\Pi\left(e^{\theta obsC}\right) > \Pi\left(e^{\theta eff}\right)$: the seller makes higher profits than if she would employ the efficient effort computed in (19). Total surplus in the efficient effort case, however, will be* higher *than under the commitment & observable type case.*

Commitment and Unobservable Type

Now we assume that the seller can still credibly commit to a certain level of effort *before* the buyers bidding decision, but the type is unobserved. This is plainly a signaling game between the seller and current and future buyers.

Lemma 3 *In this signaling game where a seller can commit to some effort level before the auction starts, a separating equilibrium doesn't exist. A pooling equilibrium in which both types of seller announce the same effort decision in both periods always exists. The maximal effort levels to which a seller of type θ can commit in both periods depend on the initial reputation μ_0 and on the number of buyers N.*

Proof. See appendix D.

The reasoning is the following: assume a separating equilibrium where a seller of type g announces e^g and a seller of type b announces e^b, and where $e^g < e^b$ w.l.o.g. Types would then be revealed right after the announcement, which would imply $\mu_1 = \mu_2 = 1$ if $e = e^g$, or $\mu_1 = \mu_2 = 0$ if $e = e^b$.

4.2 Strategic Competition for Heterogeneous Buyers

Understanding the fierceness of competition in the future is relevant in order to decide what kind of reputation building is the seller willing to embark herself in. In this section, we investigate the role of such mechanism in understanding the equilibrium behavior of buyers and sellers, as well as its potential to explain observed outcomes.

A buyer interested in the good can observe the reputation generated from past data for *all* the available sellers of that particular product. As it was mentioned before, a seller observes this very *same* information about other sellers who are competing against her. If she believes that buyers select a seller taking into account the reputational information, then other sellers reputations become payoff relevant when deciding about effort. Current beliefs will help to predict the competitor's behavior and therefore the strength of competition faced *next* period, given the updating of reputation. Sellers then start to compete *strategically*, understanding the effect of their actions not only in their absolute reputation value, but also in their "attractiveness" relative to other sellers.

Conditional on her type, and given the information that buyers consider when selecting an auction and bidding, seller j instantaneous profits in period $t = 1, 2$ are:

$$\pi_{j,t}\left(e_{j,t}^{\theta_j}, \tilde{e}_{j,t}^{\theta_j}, \tilde{e}_{-j,t}^{\theta_{-j}}, \mu_{j,t-1}, \mu_{-j,t-1}\right) = \mathbf{R}_{j,t}\left(r_{j,t}, r_{-j,t}\right) - c\left(e_{j,t}^{\theta}\right)$$

where $r_{j,t}$ was defined in equation (5), as a function of $e_j^{\theta_j}$ and μ_j. $\mathbf{R}_{j,t}\left(r_{j,t}, r_{-j,t}\right)$ is the expected revenue of seller j at time t. The expected revenue for seller j given his and his competitor's reputation is simply the expected second highest bid, given by

$$\mathbf{R}_{j,t}\left(r_{j,t}, r_{-j,t}\right) \equiv \mathbb{E}\left(b_{j,t}^{(2)}\right) = r_{j,t} H_{2j}\left(r_{j,t}, r_{-j,t}\right) \tag{22}$$

where, as before, $H_{2j}\left(r_{j,t}, r_{-j,t}\right) \equiv \mathbb{E}\left(v^{(2)}\right)$ is the expected second highest valuation in seller j's auction, given the perceived probability of a successful delivery by each seller in the current period. Expression (22) makes clear that now seller j's reputation affects her current revenues through an additional channel relative to the one seller case: the *expected valuation* of the buyers, as well as the *number* of buyers, that will decide to bid in seller j's auction, summarized by $H_{2j}\left(r_{j,t}, r_{-j,t}\right)$.

4.2.1 Expected 2^{nd} Highest Valuation in Seller j's Auction

In our setting, when there is just one seller in the market, such seller receives the same expected type of buyers irrespective of her reputation (although, as we argued, the amount that each buyer will bid does depend on the reputation). When two sellers compete, buyers need to select and bid taking into account the relative reputations. $H_{2j}(\cdot,\cdot)$, both in the first and in the second period, will now emerge *endogenously* from the auction selection and bidding equilibrium. Here we present the main features of the selection equilibrium; a more detailed description of H_{2j} is presented in appendix B.

A buyer with private valuation v^i who has the chance to bid in only one period and one auction faces the following trade-off: on the one hand bidding in the auction of the seller with the highest reputation results in a higher expected revenue upon winning but it also entails a lower probability of winning the auction. On the other hand, winning the auction of a seller with a relative lower reputation is more likely but also the expected revenue upon winning that auction is lower. This trade-off is clear from the expression (8) for the expected surplus that a buyer receives from bidding on seller j's auction. Following Bapna et al. (2010), if we restrict the analysis to symmetric selection strategies which are piecewise continuous on a buyer's type, this trade-off generates a selection equilibrium with three main features that we describe in proposition 3 below.

Proposition 3 ***One-bid selection equilibria***. *Assume, without loss of generality, that $r_j \leq r_{-j}$. When we restrict selection strategies to be piecewise continuous in buyer's type, then:*

1. *There is no pure strategy selection equilibria.*

2. *Buyers are divided into 2 type zones, with a zone threshold t_v^*:*

$$t_v^*(r_j, r_{-j}) = \sqrt[N-1]{\frac{r_j}{r_{-j}}} \tag{23}$$

3. *Buyers with type $v^i < t^*$ choose a selection strategy*

$$s^i(r_j, r_{-j}) = \left(s_j^i, s_{-j}^i\right) = \left(\frac{\sqrt[N-1]{r_{-j}}}{\sqrt[N-1]{r_j} + \sqrt[N-1]{r_{-j}}}, \frac{\sqrt[N-1]{r_j}}{\sqrt[N-1]{r_j} + \sqrt[N-1]{r_{-j}}}\right) \tag{24}$$

   while buyers with type $v^i \geq t^*$ choose a selection strategy

$$s^i(r_j, r_{-j}) = \left(s_j^i, s_{-j}^i\right) = (0, 1) \tag{25}$$

where s_j^i is the probability that buyer i assgns to bidding in auction (seller) j.

Proof. See appendix D.

Proposition 3 is a straightforward extrapolation of lemma 2 and proposition 2 in Bapna et al. (2010) to our setting. The assumptions that allows us to do this in our case are two: (i) the fact that we are focusing on short lived buyers, (ii) that use Markov strategies, mappings from the space of reputations to the space of actions. Buyers auction selection strategy depends on the perceived probability of successful delivery for all sellers, $r_{j,t}$ and $r_{-j,t}$. These probabilities are a function of reputation $(\mu_{j,t-1})_{j=1}^{2}$, effort decisions in equilibrium and the total number of bidders N. The equilibrium behavior, represented by the expected second highest bid in seller j's auction, then point out at a form of probabilistic assortative matching: buyers self-select into 2 type zones. Those in the lower type zone will mix between the two sellers, whith selection probabilities being inversely proportional to the "trustworthiness" (i.e. r) of an auction. Those on the upper type-zone bid with certainty in the more trustworthy seller.

4.2.2 First Period Competition Through Reputation Building

As in the one seller case, effort in the second period for seller j has no value ;therefore, minimum effort is still optimal: $e_{j,2}^{\theta_j} = \underline{e} \ \forall \ j, \theta_j$.

In the first period, conditional on the initial reputations $(\mu_{j,0}, \mu_{-j,0})$, each seller j of type θ is faced with the following problem

$$\max_{e_{j,1}^{\theta_j}} \left\{ r_{j,1} H_{2,j}\left(r_{j,1}, r_{-j,1}\right) - c\left(e_{j,1}^{\theta_j}\right) + \delta \mathbb{E}\Big(r_{j,2} H_{2,j}\left(r_{j,2}, r_{-j,2}\right) - c\left(\underline{e}\right) \Big| e_{j,1}^{\theta_j}, \theta_j \Big) \right\} \quad (26)$$

s.t.

$$e_{j,1}^{\theta_j} \in [\underline{e}, 1]$$

Here, $r_{j,t} = P\left(y_{j,t} = S | \boldsymbol{\mu_{t-1}}\right)$ is the probability of success attributed to seller j at the beginning of period t.

Given current beliefs $\boldsymbol{\mu}_t$, beliefs in $t+1$ will depend on $y_{j,t}$ and the predicted effort for each type of seller. As in previous sections, we denote $\mu_j^S(e_j)$ and $\mu_j^F(e_j)$ the updated beliefs after $y_{j,t}=S$ and $y_{j,t}=F$ respectively. Denote seller's j profit in period 2 as $\pi_{j,2}^{y_{j,1}y_{-j,1}}$, conditional on his and his competitor outcome of the first period $y_{j,1}$ and $y_{-j,1}$. In addition, we denote $r_{j,2}^{y_{j,1}}$ the anticipated probability of success of seller j in the second period, given the outcome of his transaction in the first period $y_{j,1}$: $r_{j,2}^{y_{j,1}} = \left(\mu_{j,1}^{y_{j,1}} g\tilde{e}_{j,2}^{g} + \left(1-\mu_{1,1}^{y_{j,1}}\right) b\tilde{e}_{j,2}^{b}\right)$.

Given the (conditional) independence of outcomes, the expected profit of seller j at time 2, given her type, effort and perceived effort from her competitor is

$$\begin{aligned}
E\left[\pi_{j,2}|e_{j,1}^{\theta_j},\theta_j\right] &= e_{j,1}^{\theta_j}\theta r_{-j,1}\pi_{j,2}^{SS} + e_{j,1}^{\theta_j}\theta_j\left(1-r_{-j,1}\right)\pi_{j,2}^{SF} \\
&+ \left(1-e_{j,1}^{\theta_j}\theta_j\right) r_{j,1}\pi_{j,2}^{FS} + \left(1-e_{j,1}^{\theta_j}\theta_j\right)\left(1-r_{-j,1}\right)\pi_{j,2}^{FF}
\end{aligned}$$

Again, effort best replies to all the other seller's effort (or to their current reputation) are characterized by the FOCs of the problem:

$$c'\left(e_{j,1}^{\theta_j}\right) = \theta_j\delta\Big(r_{-j,2}\left[\pi_{j,2}^{SS}-\pi_{j,2}^{FS}\right] + \left(1-r_{-j,1}\right)\left[\pi_{j,2}^{SF}-\pi_{j,2}^{FF}\right]\Big) \tag{27}$$

When deciding about effort after auctions in the first period have closed, the timing and lack of commitment from sellers implies that they will only compete in *reputations* for future revenues $\boldsymbol{R}_2$. In other words, an announcement to a certain effort level is ruled out by a typical time inconsistency problem. Proposition 3 reveals why the revenues in period 2 for seller j are a function period-2 buyers' beliefs about the probability of success of both sellers; these probabilities, in turn, arise through the updating of reputation after the effort decisions and outcome in period 1. Therefore, seller's j strategic behavior will weight the current cost of effort against the (discounted) potential future revenue reaps from her competitor; this is the RHS term between parenthesis in eq. (27).

In equilibrium, revenue reaps will be determined by the interaction of three components: (i) the updating function; (ii) the valuation threshold $t_v^*(r_1,r_2)$; and (iii) the bidding probabilities $s_j^i(r_j,r_{-j})$. The current public information about her type and her competitors type is therefore payoff relevant for a seller since it helps predict her competitor's action (effort) in the current period, which is also targeted at influencing buyer's beliefs next period. More precisely, the reputations at the beginning of period 1 (i.e. $\boldsymbol{\mu_0}$) allows sellers to understand the marginal effect of effort when there is competition.

Before characterizing the equilibrium(s), we show that the primitives of the game generate effort decisions that are monotonic in types, or what is usually known as single-crossing condition within a revealed preference argument. We state this in Lemma 4 below.

Lemma 4 Single-Crossing condition: *Let* $e_{j,1}^{\theta_j} : [0,1]^2 \times \Theta \longrightarrow [0,1]$ *be an effort strategy for seller* $j = 1, 2$, *as a function of her type* $\theta_j \in \Theta$ *and reputations* $\boldsymbol{\mu} \equiv \{\mu_j, \mu_{-j}\}$. *Then, given the multiplicative technology* $P(y_j = S|\theta, e) = \theta e$, *it is the case in equilibrium that* $e_j^g(\boldsymbol{\mu}) \geq e_j^b(\boldsymbol{\mu})$, $\forall \boldsymbol{\mu} \in [0,1]^2$, $\forall j$.

Proof. See appendix D.

It turns out that, similar to the one seller case, there is a unique equilibrium in (non-stationary) Markov strategies. This equilibrium has two interesting (and somewhat counterintuitive) properties regarding the comparative statics on a competitor's prior reputation and the total number of sellers in the market. Propositions 4 and 5 below state these features rigorously.

Proposition 4 *An equilibrium of the two periods game in which two sellers of unobservable are competing for heterogenous buyers exists and is unique.*

Proof. See appendix D.

Proposition 5 ***Characterization of the equilibrium*** *In the Markov equilibrium where two sellers compete during two periods through second price auctions,*

1. *Seller j's ($j = 1, 2$) effort decision in period 1 is still non-monotonic in her own current (prior) reputation $\mu_{j,0}$, for $\theta \in \{g, b\}$:* $\frac{\partial e^{\theta}_{j,1}(\mu_{j,0},\mu_{-j,0})}{\partial \mu_{j,0}} > 0$ *for* $\mu_{j,0} < \mu^*$ *and* $\frac{\partial e^{\theta}_{j,1}(\mu_{j,0},\mu_{-j,0})}{\partial \mu_{j,0}} < 0$ *for* $\mu_{j,0} > \mu^*$*. The threshold* μ^* *is a function of* $\mu_{-j,0}$ *and the different parameters of the game (in particular, of* g *and* b*).*

2. *For any* $\left(\mu_{j,0}, \mu_{-j,0}\right) \in [0, 1]^2$ *and type* $\theta \in \{g, b\}$*, seller j's effort decision in period 1 is monotonically decreasing in her competitor's prior reputation* $\mu_{-j,0}$ *("intensity" of competition):* $\frac{\partial e^{\theta}_{j,1}(\mu_{j,0},\mu_{-j,0})}{\partial \mu_{-j,0}} < 0$.

3. *An increase in the total number of buyers in the market from* N *to* N'*, with* $N' > N$ *and* $N, N' \geq 2$ *has two effects:*

   (a) It ***reduces the impact*** *of competitor's period-1-reputation,* $\mu_{-j,0}$*, on the incentives to build one's own reputation:* $\left|\frac{\partial e^{\theta}_{j,1}(\mu_{j,0},\mu_{-j,0};N)}{\partial \mu_{-j,0}}\right| > \left|\frac{\partial e^{\theta}_{j,1}(\mu_{j,0},\mu_{-j,0};N')}{\partial \mu_{-j,0}}\right|$ *; and*

   (b) for any $\left(\mu_{j,0}, \mu_{-j,0}\right) \in [0, 1]^2$*, it* ***increases*** *the equilibrium effort* ***level*** *of both types of seller* j*,* $j = 1, 2$*:*
   $e^{\theta}_{j,1}\left(\mu_{j,0}, \mu_{-j,0}; N\right) < e^{\theta}_{j,1}\left(\mu_{j,0}, \mu_{-j,0}; N'\right)$ *for* $\theta \in \{g, b\}$.

Proof. See appendix D.

Discussion

Proposition 5 presents three interesting comparative statics results related to the impact of period-1 *strategic competition*, on reputation building. The first result states that, for a *fixed* prior belief about the competitor's type, effort in period 1 is non-monotone and concave in a seller's own prior reputation. This implies that the reputation mechanism discussed for the case of one seller is still in place. The second and more important result shows that a higher prior (current) reputation of a seller's competitor leads the seller to employ lower effort c.p. Third, there are two channels through which the size of the market influences sellers' behavior.

Below, we discuss the latter two features, since the first one has been already discussed for the case of 1 seller. In order to explain the intuition behind these, we will refer to figures 5, 4 and figure 6, in which we present, respectively, the first period equilibrium effort, the behavior of the equilibrium valuation threshold $t_v^*(r_j, r_{-j})$, and the expected second highest valuation in a seller's auction for any pair of reputation values of the two sellers.

Comparative statics result 2 is rather counterintuitive. Figure 5 plots period-1 effort levels and profits; it considers the case in which sellers can be "very trustworthy" or "barely trustworthy" (i.e. $g = 0.9$ and $b = 0.1$), both for a small and a large number of bidders ($N = 5$ in the left column and $N = 50$ in the right column). The first row presents seller 1 effort levels as a function of her own reputation, for a fixed value of seller reputation; the second row plots seller 1 effort as a function of competitor reputation, for a fixed value of her own reputation. One the one hand, when introducing even minimal competition (i.e. adding a seller with a minimal reputation, but maintaining the number of buyers fixed) the expected level of revenues for a seller of any given reputation decreases, since as the pool of buyers that will self-select into the seller's auction is now (weakly) smaller. This is not surprising, but it is not what drives *marginal* effort decisions under competition. On the other hand, one would expect that the introduction of competition would, for certain regions of the relative reputation space, rise the *marginal* benefits from exerting effort. When the reputation of the competitor increases, effort increases would intuitively be generating (with higher probability) not only a better absolute reputation next period but also a base of bidders with better valuations, counterbalancing the improved competitor's reputation[12]. However, the opposite seems to be happening: when the prior reputation of a seller's competitor increases, the incentives for the seller to exert effort *decrease*.

FIGURE 5 ABOUT HERE

The explanation behind such decreasing incentives as the opponent's reputation improves, is to be found in how these help a seller predict her competitor's effort and its effect on future relative reputations. Ultimately, the latter will determine, through buyers behavior, period-2 revenues.

[12] This analysis obviously does not take into consideration entry or exit decisions.

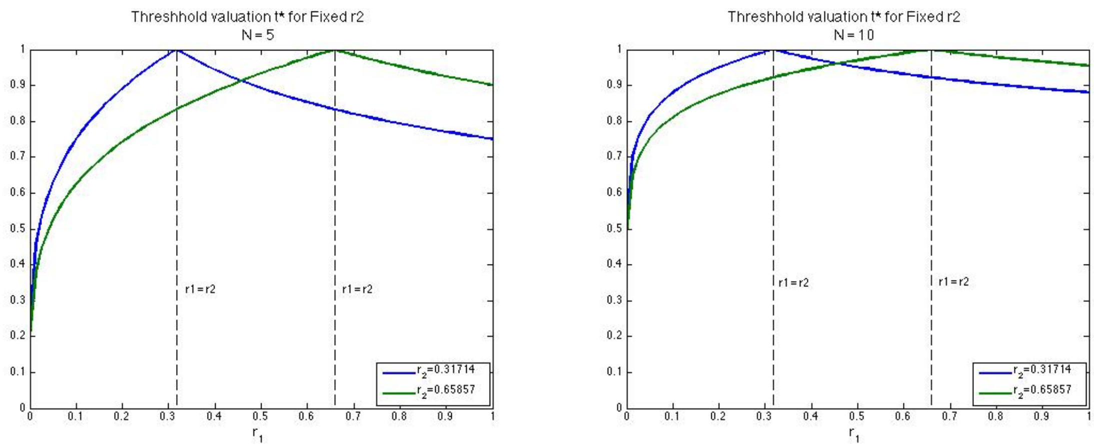

Figure 4: Equilibrium behavior of $t_v^* (r_j, r_{-j})$, $N = 5$ (left) and $N = 15$ (right). Fixed values of r_{-j}

Recall that in the second period, effort is minimum: $e_{j,2}^{\theta} = \underline{e}\ \forall j, \theta$. Therefore, the perceived probability of a successful delivery $r_{j,2}$ will be completely determined from the updating of reputation, $\mu_{j,2}^{y}$. Given the pair of success probabilities, the behavior of the threshold valuation $t_v^* (r_j, r_{-j})$ and the bidding probabilities $s^i (r_j, r_{-j})$ as defined in proposition 3 determine the behavior of revenues for both sellers. Figure 4 plots the valuation threshold t^* as a function of seller 1 success probability r_1, for two values of r_2 (blue and green lines). When seller 1 has relatively low reputation in the last period (to the left of the dashed lines in both plots), an increase in her reputation increases t^* towards 1 (where $r_1 = r_2$). More importantly, an increase in the competitor's reputation (making seller 1 relative reputation even smaller) is translated into a *decrease* in the slope of $t_v^* (r_1, r_2)$. This implies a *smaller marginal return* of her own reputation, measured in terms of the type of buyers that will decide to bid in her auction. A similar behavior occurs when $r_1 > r_2$, i..e., when seller 1 has a relatively high reputation. For this region, an increase in r_1 implies that $t_v^* (r_1, r_2)$ will decrease and fewer buyers will decide to bid with probability ¿0 in seller 2, although the slope increases (decreases in absolute value) for higher values of r_2. These characteristics of t^* are the kernel of the instantaneous revenues of a seller (expected second highest bid).

The above properties from the selection equilibrium, combined with the way in which reputations are updated, shape the strategic competition in the first period. In the third row of figure 5 we plot the expected revenue function in equilibrium (incorporating equilibrium effort decisions predicted by buyers; the key characteristic is that such function it exhibits decreasing differences. The main implication is that a seller's marginal expected revenue increases when his competitor's reputation $\mu_{-j,0}$, and therefore $r_{-j,0}$, decrease. This reasoning can be seen in the right hand side of the FOC 27 for seller j and type θ: the lower is the public reputation of one's competitor, the more weight is put on the second term of the sum on the RHS of 27, which, given the decreasing differences property of the expected revenues function, is also the bigger term. In other words, when competitor's reputation $\mu_{-j,0}$ and respectively his success probability r_{-j} are low, seller j has higher incentives to exert higher effort to attain a success and maintain/improve his reputation.

The number of bidders impacts the future profits through two channels: on the one hand, in line with Bapna et al. (2010) analysis, a higher number of bidders reduces the effect of relative reputations on buyers' auction selection strategies, and on the other it leads to an increase in the expected price in any single auction due to a larger pool of bidders available. Expected higher future profits trigger the seller to employ higher effort in the present. When the number of bidders increases, their assortative bidding behavior converges towards perfectly randomizing between sellers. This effect, as argued in the previous section, is due to the fact that a high total number of auction participants reduces the chances of each of them to win any particular auction and reduces the expected benefit in the event of winning an auction. Hence, for high competition among bidders (high N), their optimal strategy prescribes them to perfectly randomize between sellers. The change in buyers' bidding behavior as we vary N has a significant impact on sellers' expected profits and hence on their optimal effort. As the number of bidders increases, the relative reputations ranking becomes irrelevant as both sellers will receive bids equally probable from all bidders. In addition to the vanishing assortative bidding, ahigher number of bidders in each auction c.p. increases the chances that the bidders have higher valuations which leads to higher expected profits.

5 Final Remarks

On-line markets, loan securitization (or more broadly, asset backed securities) markets or debt markets are characterized by the intrinsic asymmetry of information regarding seller (or borrower) trustworthiness, as well as the effort that an originator, a seller or a borrower put in order to monitor a loan, deliver a good or efficiently manage a project. In addition, agents in such markets are usually non-anonymous, which means that reputation building can potentially arise as a way to tackle the adverse selection or moral hazard problems.

We have shown that, contrary to the common intuition about the effect of (oligopolistic) competition in a general market setting, a seller' incentives to build a reputation through effort in order to deliver and complete a transaction decrease when competition occurs within a 2nd price auction environment and when the reputation of the competitor increases. This is an important insight, not only from a theoretical point of view, but also from a more practical perspective, since it allow us to shed light on the usefulness of reputation measures as a way to solve the "lemons problem" in on-line markets.

Crucially, in on-line markets such as ebay the public information structure available to sellers and buyers is the same as the one available to us researchers (except, perhaps, for potential email correspondence between parties). This implies that predictions made by buyers about the behavior or characteristics of a particular seller necessarily use public information which we can observe. This, together with the dynamic game framework could allow us to study quantitatively two identification problems related to the reduced-form estimation of the impact of seller reputation measures on revenues, and that have been omitted by recent empirical papers. Biases in the estimation of this elasticity might arise when reputation building and strategic interactions are not controlled for, since both reputations can be correlated in equilibrium. In other words, the omission of strategic behavior can bias the estimate of the reputation elasticity of revenues. Nonetheless, our theoretical results also hint that when there is a reasonable number of buyers in the market, the consequences of omitting information about other sellers could be milder.

6 Figures

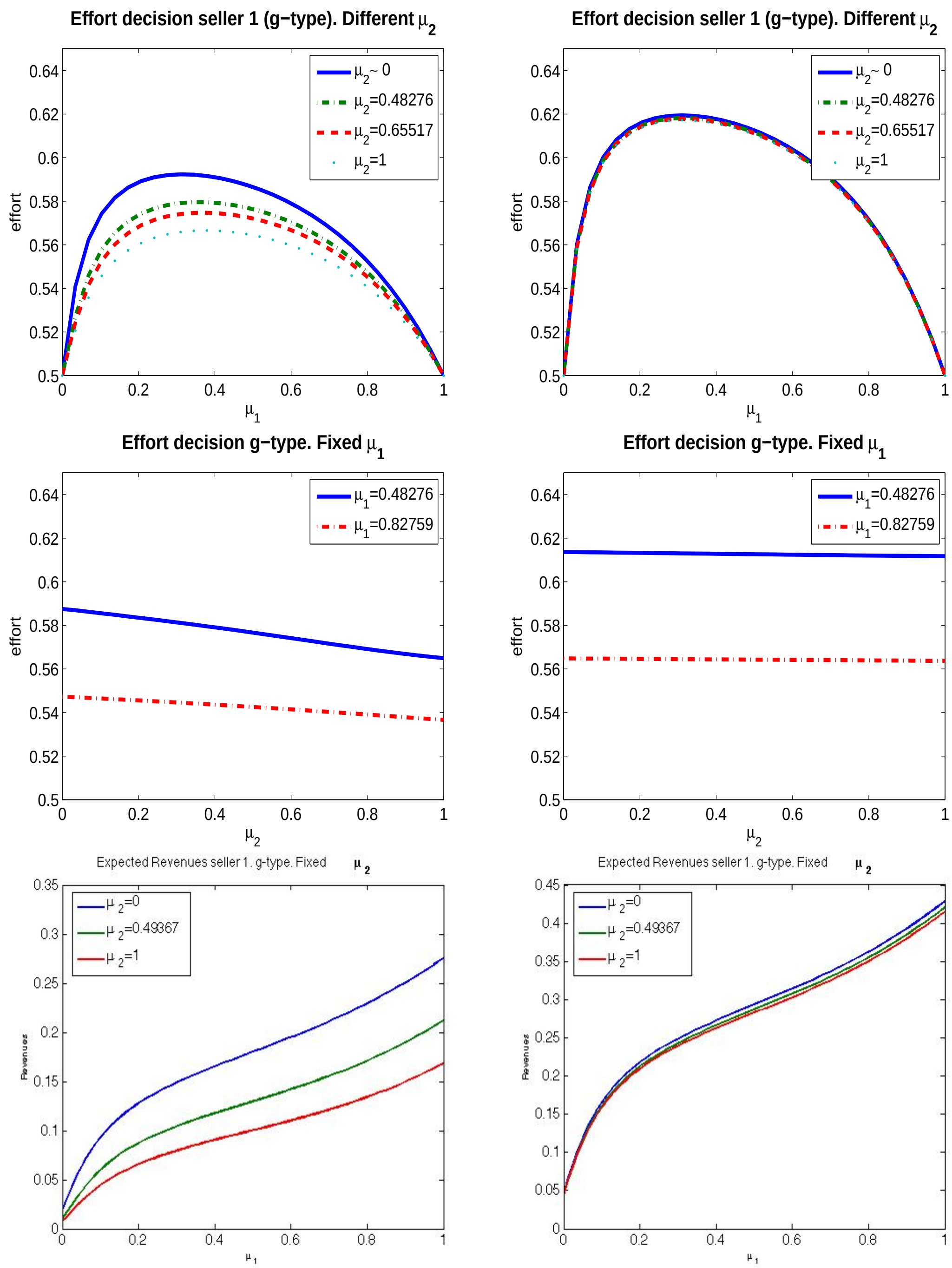

Figure 5: Equilibrium efforts in period 1 (top, middle) and PV profits (bottom). Left column: $N = 5$ Right column: $N = 50$

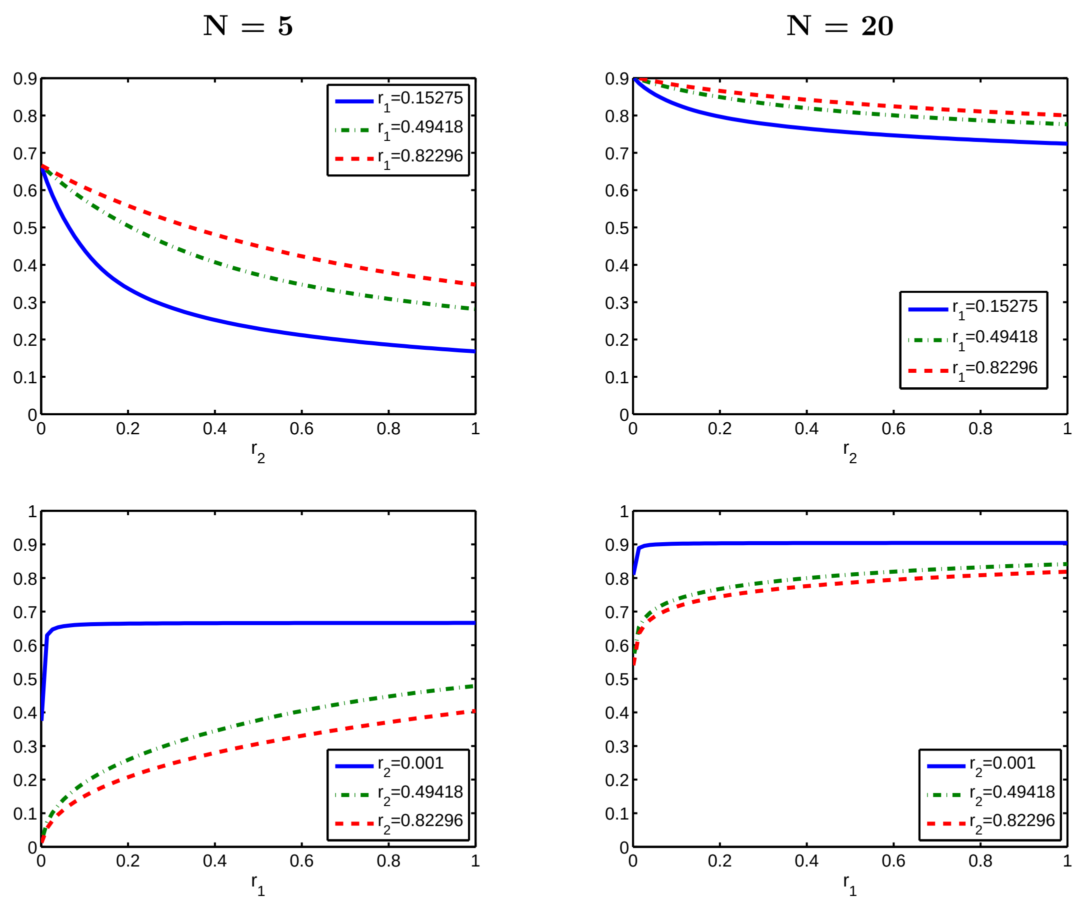

Figure 6: Expected second highest bid on **seller 1 auction**: fixed r_1 (top) and fixed r_2 (bottom). Left: $N = 5$ Right: $N = 20$

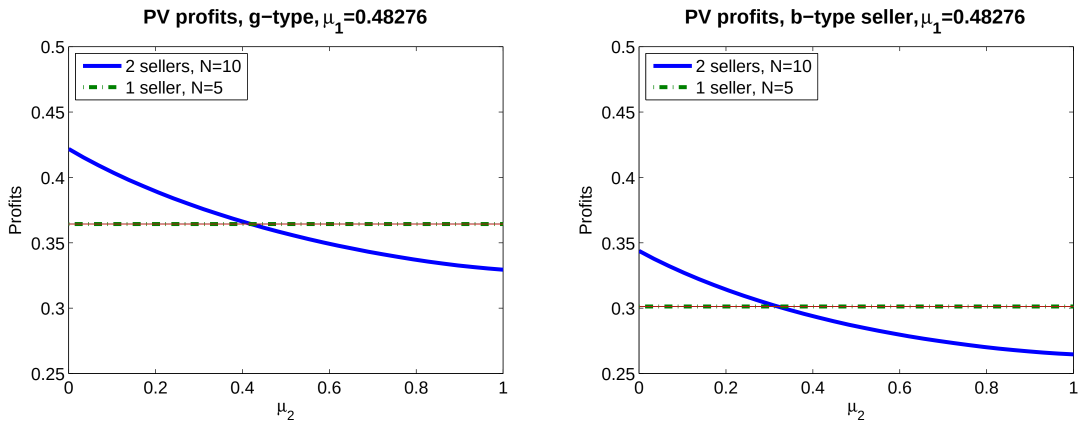

Figure 7: Present Value Profits (PV) for one seller, as a function of the competition's reputation, in the (i) one seller game, (ii) 2-sellers game. Left: "good" sellers (g-type). Right: "bad" sellers (b-type). For a proper comparison, the number of buyers in the 2-sellers case is 2x the number of buyers in the 1-seller case. In the figures: N=5 for the 1-seller case and N=10 for the 2 sellers case.

References

Adlakha, S., R. Johari, and G. Y. Weintraub (2010). Equilibria of Dynamic Games with Many Players: Existence, Approximation, and Market Structure. *Management Science* (April), 71.

Akerlof, G. A. (1970, August). The Market for 'Lemons': Quality Uncertainty and the Market Mechanism. *The Quarterly Journal of Economics 84*(3), 488–500.

Athey, S. (2001, July). Single Crossing Properties and the Existence of Pure Strategy Equilibria in Games of Incomplete Information. *Econometrica 69*(4), 861–89.

Bapna, R., C. Dellarocas, and S. Rice (2010, July). Vertically Differentiated Simultaneous Vickrey Auctions: Theory and Experimental Evidence. *Management Science 58*(7).

Cabral, L. and A. Hortaçsu (2010). The Dynamics of Seller Reputation: Evidence from eBay. *The Journal of Industrial Economics 58*(1), 54–78.

Cripps, M. W., G. J. Mailath, and L. Samuelson (2004). Imperfect Monitoring and Impermanent Reputations. *Econometrica 72*(2), 407–432.

Doraszelski, U. and A. Pakes (2007, 00). *A Framework for Applied Dynamic Analysis in IO*, Volume 3 of *Handbook of Industrial Organization*, Chapter 30, pp. 1887–1966. Elsevier.

Ericson, R. and A. Pakes (1995). Markov-Perfect Industry Dynamics: A Framework for Empirical Work. *The Review of Economic Studies 62*(1), pp. 53–82.

Escobar, J. F. (2011). Equilibrium Analysis of Dynamic Models of Imperfect Competition. *Working paper. University of Chile*.

Holmstrom, B. (1999). Managerial Incentive Problems: A Dynamic Perspective. *The Review of Economic Studies 66*(1), 169–182.

Horner, J. (2002, June). Reputation and Competition. *American Economic Review 92*(3), 644–663.

Jullien, B. and I.-U. Park (2011, March). Seller Reputation and Trust in Pre-Trade Communication. The Centre for Market and Public Organisation 11/272, Department of Economics, University of Bristol, UK.

Kreps, D. and R. Wilson (1982, August). Reputation and Imperfect Information. *Journal of Economic Theory 27*(2), 253–279.

Mailath, G. J. and L. Samuelson (2001, April). Who Wants a Good Reputation? *Review of Economic Studies 68*(2), 415–41.

Mason, R. and A. Valentinyi (2007, October). The existence and uniqueness of monotone pure strategy equilibrium in bayesian games. Discussion paper.

Resnick, P., R. Zeckhauser, J. Swanson, and K. Lockwood (2006). The Value of Reputation on eBay: A Controlled Experiment. *Experimental Economics 9*, 79–101.

Tadelis, S. (2002, August). The Market for Reputations as an Incentive Mechanism. *Journal of Political Economy 110*(4), 854–882.

Vial, B. (2010). Walrasian Equilibrium and Reputation under Imperfect Public Monitoring. *The B.E. Journal of Theoretical Economics 10*(1), 23.

Vial, B. and F. Zurita (2012, July). Competing Reputations under Free Entry. Universidad de Chile Working Paper.

APPENDIX

A Feedback Systems in On-line Markets: Some Examples

The feedbacksystem is the backbone of online markets in which anonymous parties are transacting. Its role is to facilitate transactions by inducing trust in interactions in which traditional contractual agreements are not always enforceable. A feedback mechanism, otherwise called *reputation system*, is a public record of a trader's past transactions. Revealing an agent's past actions will affect the future behavior of the entire comunity towards that agent. It can hence discipline traders' behavior and provide them with incentives not to missbehave in the short-run.

In the early 2000, eBay was by far the most successful on-line market with almost fifty million users and twelve million auctions run daily. The success of the platform is due moslty to its feedback system. eBay's reputation system has a simple and for all users accessible design. At the end of each transaction, both parties can evaluate each other using ratings from a tripartite scale- positive (1), neutral (0) and negative (-1) - and an optional short comment. Following the argument in Cabral and Hortaçsu (2010), according to which neutral grades are rather perceived as negative ratings, we devise our model with a binary rating system. We assume that traders in our model thuthfully report the outcome of each transaction as either a "Success"(S) or a "failure"(F). Under the assumption that bidders are interested in purchasing one object only and hence their aim is not to build a reputation, we restrict our attention to the sellers' track record. The aggregate measure used on eBay as a reputation indicator consists of the number of positive net of negative ratings. The bidders in our model are Bayesian updaters. The reputation measure they attach to a particular seller is their belief that the seller they are facing is of a certain type.

eBay's feedback system doesn't distinguish between feedbacks received from sellers or buyers. Cabral and Hortaçsu (2010) documents that in general, traders start their career as buyers and accumulate a number of positive feedbacks. With an established reputation, they then start their seller's career. For simplicity, we assume that traders can be either sellers or buyers. Sellers in our model enter the game with a given reputation and build it gradually.

Even though eBay's feedback system promotes cooperation, it suffers from a couple of severe drawbacks. The free entry and exit for all market participants opens the door for opportunistic behavior. Sellers can choose to exit the market and re-enter under a new identity at basically no cost. Cabral and Hortaçsu (2010) document that before exiting the market, sellers "milk" their reputation. In the benchmark version of our model we assume that sellers don't have the option to exit the market. In a further extension of our model we investigate the case in which sellers can exit the market once their reputation has fallen below a threshold and can opt for re-entry against paying a fixed cost.

B Buyer's Selection Strategy and Sellers' Expected Revenues

Buyer's Selection Strategy

In what follows, we describe the main features that characterize the buyer's optimal selection strategy, and the resulting expected type of the second highest bidder in seller's j auction.

The *ex-ante* utility of a buyer with valuation v^i bidding in j's auction is defined as following:

$$\pi^i(v^i) = \sum_{n=1}^{N} Pr[n \text{ other bidders choose auction} j] \times Pr[\text{all } n \text{ other bidders have types} \leq v^i]$$
$$\times r_j \times (v^i - E[\text{2nd highest type in j's auction} \mid \text{highest type} = v^i \wedge \exists\, n \text{ other bidders}]) \tag{28}$$

Following Bapna et al. (2010), we introduce a series of measures that will prove useful in defining the equilibrium results. Given the auction selection strategy of a bidder $s : [0,1] \times [0,1]^J \Rightarrow [0,1]^J$, the following function defines the probability that a buyer is of type v^i or lower *and* bids in seller's j auction.

$$Q_j(v^i|s) = \int_0^{v^i} s^i_j(u) f(u) du, \quad j = 1, \ldots, J$$

The expression $Q_j(1|s)$ stands for the probability that a random buyer bids in j's auction, whereas $1 - Q_j(1|s) + Q_j(v^i|s)$ is the probability that a random buyer is not of a higher type than v^i *and* bids in j's auction.

It results that the above defined expected profit of a buyer with valuation v^i bidding in j's auction can be written as: $\pi^i(v^i) = \int_0^{v^i} (1 - Q_j(1|s) + Q_j(x|s))^{N-1} dx$

Further we provide a short description of buyers' equilibrium strategies as outlined Proposition 3 in Dellarocas Bapna et al. (2010).

In the general set-up with J sellers in which we have assumed a complete order of their reputations, let there be an L defined as the lowest integer $2 \leq L \leq J - 1$ for which $r_{L+1} < \left(\frac{L-1}{\sum_{j=1}^{L} \sqrt[N-1]{\frac{1}{r_j}}} \right)^{N-1}$. If the integer does not exist, then $L = J$. The unique one-bid auction selection equilibrium is described as following:

1. Buyers are divided into L zones according to their types. Let t_z $z = 0, 1, \ldots, L$ with $t_0 = 1$, $t_L = 0$ be the zone delimiters. Buyers whose types satisfy $t_z < t \leq t_{z-1}$ belong to zone z.

2. Zone z buyers randomly chose among sellers $j = 1, \ldots, z$ with corresponding selection probabilities:

$$s_{zj} = \frac{\sqrt[N-1]{\frac{1}{r_j}}}{\sum_{j=1}^{z} \sqrt[N-1]{\frac{1}{r_j}}}$$

3. If $L < J$, then sellers $L + 1, \ldots J$ are never chosen by any buyer

4. Zone delimiters t_z $z = 0, 1, \ldots, L-1$ are the solution to the following equations:

$$F(t_z) = \left(\sqrt[N-1]{r_{z+1}} \sum_{j=1}^{z} \sqrt[N-1]{\frac{1}{r_j}} \right) - (z - 1)$$

B.0.1 Sellers' Expected Revenues

Lemma 3 in Dellarocas Bapna et al. (2010) provides the closed-form solution of seller's j expected revenue. In the auction of a seller with reputation r_j, where we assume w.l.o.g. $r_1 \geq \ldots \geq r_j \geq \ldots \geq r_M$, the expected type of the second highest bidder given their equilibrium strategies is given by:

$$H_{2,j}(r_1, ..., r_J) = 1 + (N-1) \int_0^1 (1 - Q_j(1) + Q_j(t))^N \, dt - N \int_0^1 (1 - Q_j(1) + Q_j(t))^{N-1} dt$$

The function $H_{2,j}(r_j, r_{-j})$ is increasing in seller's j reputation and decreasing in his competitors' reputations.

When reputations are equal $r_1 = r_2 = \ldots = r_J$, the expected type of the second highest bidder is the same in each auction and equals:

$$H_2(r_j, r_{-j}) = 1 + \frac{N-1}{J^N} \int_0^1 (M - 1 + F(t))^N dt - \frac{N}{M^{N-1}} \int_0^1 (M - 1 + F(t))^{N-1} dt$$

In the case of one seller in the market, the expected type of the second highest bidder is constant throughout the periods and equal to $H_2(N) = \frac{N-1}{N+1}$, assuming the monopolist faces the same number of buyers N every period whose valuations are drawn from the same distribution. In the case of only one seller or when all sellers have the same reputation, they profit vary linearly with their reputation.

Two Sellers

For the case of two sellers, further called i and j, our results are summarized in the following section. We assume that the reputations of the two sellers satisfy the following relation $r_{-j} > r_j$.

- *Assortative Matching:* there exists a threshold type $t_{j,-j}$ such that bidders with valuations higher than $t_{j,-j}$-bidders of zone $-j$- bid their expected valuation in the auction of the seller with the highest success probability, which we assume to be r_{-j}. Buyers whose types are below this threshold have a mixed optimal auction selection strategy that places a higher weight in choosing the auction of the seller with the lowest reputation r_j. Under the assumption of a uniform distribution of buyers' valuations, the threshold value and strategy selection vector take the following form:

$$
\begin{aligned}
t_{j,-j}(r_{-j}, r_j) = \sqrt[N-1]{\frac{r_j}{r_{-j}}} \\
s_{-j} = (s_{-j,-j}, s_{-j,j}) = (1, 0) \\
s_j = (s_{j,-j}, s_{j,j}) = \left(\frac{\sqrt[N-1]{r_j}}{\sqrt[N-1]{r_j} + \sqrt[N-1]{r_{-j}}}, \frac{\sqrt[N-1]{r_{-j}}}{\sqrt[N-1]{r_j} + \sqrt[N-1]{r_{-j}}} \right)
\end{aligned}
$$

- As a result, the seller with the highest reputation, r_j, expects to get more bids and from buyers with higher valuations. The expected second highest valuation in his auction, henceforth called $H_{2,j}$ is given by

$$
\begin{aligned}
H_{2,-j}(r_{-j}, r_j) &= 1 + (N-1)\int_0^1 (1 - Q_{-j}(1) + Q_{-j}(t))^N \, dt - N\int_0^1 (1 - Q_{-j}(1) + Q_{-j}(t))^{N-1} dt \\
&= \frac{N-1}{N+1}\left(s_{j,-j}^N \left((1+t_{j,-j})^{N+1} - 1\right) + 1 - t_{j,-j}^{N+1}\right) - s_{j,-j}^{N-1}\left((1+t_{j,-j})^N - 1\right) + t_{j,-j}^N;
\end{aligned}
$$

- The seller with the relatively lower reputation, r_i, expects to get a lower number of bids than seller j and also from buyers with valuation from the low part of the distribution. The expected second highest type in i's auction is:

$$
\begin{aligned}
H_{2,j} &= 1 + (N-1)\int_0^1 (1 - Q_j(1) + Q_j(t))^N \, dt - N\int_0^1 (1 - Q_j(1) + Q_j(t))^{N-1} dt) \\
&= t_{j,-j} + \frac{N-1}{N+1}(1 - s_{j,-j})^N \left((1+t_{j,-j})^{N+1} - 1\right) - (1 - s_{j,-j})^{N-1}((1+t_{j,-j})^N - 1)
\end{aligned}
$$

C Informative signals and the dynamics of beliefs

The way in which buyers (and other sellers) update their beliefs about the seller's type (trustworthiness) depends on the "informativeness" of the *signal* they receive. In our case, the signal is simply the realization of a transaction. Observing a success (S) or a failure (F) provides the buyer with information about the type of the seller (although both are unobservable, buyers and sellers *understand* the way in which other sellers behave *in equilibrium* and therefore they acknowledge the effort decisions by the different seller types. If the probability of an outcome, conditional on effort, is relatively "flat" in types, meaning that in equilibrium it does not change much with the type of a seller, then we say that the outcome is not an informative signal.

Along the previous lines, therefore, if we observe within the model that the updating of beliefs after a failure is much more gradual than after a success, this might be due to the characteristics of the conditional density of outcomes (conditional on effort): for certain parameter values, what might be happening is that the probability of a failure is relatively high (note that such probability is an equilibrium object which depends on the effort level). This implies that observing a success provides much more information about the type of the seller than observing a failure.

Lets see this with an example. In our model, we have that the probability of a success, conditional on effort and type is

$$p\left(r = s|e^{*}, \theta\right) = e \cdot \theta$$

Lets consider as an example the benchmark case, in which $\theta \in \{0.1, 0.9\}$. When we solve the model, for a particular cost function we obtain an optimal effort level $e^{*} \in [0.1, 0.15]$. This implies that the probability of a success, conditional on type is $p^{s}_{\theta=g} \in [0.09, 0.135]$ for the good type, and $p^{s}_{\theta=b} \in [0.01, 0.015]$ for the bad type.[13] The "likelihood ratio" *for a good type* is given by

$$LR_{\theta=g} \equiv \frac{p\left(r = s|\theta = g\right)}{p\left(r = s|\theta = b\right)} = 9$$

On the other hand, the conditional probabilities of a failure are $p^{f}_{\theta=g} \in [0.865, 0.91]$ and $p^{f}_{\theta=b} \in [0.985, 0.99]$ for good and bad types respectively. In this case, the likelihood ratio *for a bad type* is given by

$$LR_{\theta=b} \equiv \frac{p\left(r = f|\theta = b\right)}{p\left(r = f|\theta = g\right)} \in [1.088, 1.138]$$

The above implies that a success is much more informative about a good type than a failure about a bad type, and therefore gives rise to an asymmetry in the updating of beliefs.

[13] This is not entirely correct, since the optimal effort functions are different for good and bad types. However, with lower effort for bad types, the argument beow would be even more extreme.

D Proofs

D.1 Proof of Lema 3

- **Separating Equilibria**

  In a separating equilibrium with commitment, a seller of type θ will announce e_1^θ in the first period and e_2^θ in the second period such that $e_1^g \neq e_1^b$ and $e_2^g \neq e_2^b$. In a separating equilibrium, upon the announcement of the policy any type would like to commit to, the type of the seller is immediately inferred and the reputation of the seller becomes degenerate.

  The problem for each type of seller would then be the following:

$$\max_{(e_1^\theta, e_2^\theta) \in [0,1]^2} \Pi^\theta = \theta e_1^\theta H_2 - c\left(e_1^\theta\right) + \delta\Big(\theta e_2^\theta H_2 - c\left(e_2^\theta\right)\Big) \tag{29}$$

  The FOCs are given by

$$\begin{aligned} c'\left(e_1^{\theta *}\right) &= \theta H_2 \\ c'\left(e_2^{\theta *}\right) &= \theta H_2 \end{aligned}$$

  implying $e_1^{\theta *} = e_2^{\theta *}$. However, unless the cost function is such that the profits of the two types are equal, $\Pi^g\left(e^{g*}\right) = \Pi^b\left(e^{b*}\right)$, one of the two types will have incentives to deviate and announce an effort level equal to the one from the other type. Thus, the separating equilibrium would break down.

- **Pooling equilibria**

  Pooling Equilibrium in the stage game

  We start our analysis by investigating the existence of a pooling equilibrium in the stage game. This correspond to analyzing the second period of our 2-period game.

  Definition 8 *A pooling equilibrium of the stage game is a Perfect Bayesian Equilibrium in which each type of agent chooses the same announcement in* *equilibrium:* $e_2^g = e_2^b = e_2^*$ *such that* $Prob(\theta = g|e_2^*) = \mu$ *and* $Prob(\theta = b|e_2^*) = 1 - \mu$ *and the expected amount bid by a principal with valuation* v^i *is* $b(v^i) = v^i(\mu g + (1-\mu)b)e_2^*$

The expected equilibrium stage profits of a monopolist in a pooling equilibrium are given by the following expression:

$$\pi(e_2^*) = (\mu g + (1-\mu)b)e_2^* H(N) - c(e_2^*) \tag{30}$$

where $H(N) = \frac{N-1}{N+1}$ is the expected second highest bid in the auction of the monopolist. To characterize the set of pooling equilibria, we need to specify first how beliefs are formed out of equilibrium. Suppose now that, whenever a seller deviates from e_2^*, buyers will attach probability 1 of him being a bad type, i.e. $\mu = 0$. Whenever a seller deviates from e_2^* and chooses to announce $\tilde{e}_2$ instead, his expected deviation profits are described by:

$$\pi(\tilde{e}_2) = b\tilde{e}_2 H(N) - c(\tilde{e}_2) \tag{31}$$

Further we compute a monopolist's maximum attainable profits under deviation. The effort level that maximizes the deviation profits of the monopolist can be computed from the FOC:

$$c'(\tilde{e}_2^*) = bH(N) \tag{32}$$

and equals:

$$\tilde{e}_2^* = c'^{-1}(bH(N)) \tag{33}$$

The maximum attainable profit under deviation is given by $\pi(\tilde{e}_2^*)$. Further, we assume that when deviating, the seller will choose $\tilde{e}_2^*$.
We investigate now what efforts can be sustained in a pooling equilibrium. Incentive compatibility requires that $\pi(e_2^*) \geq \pi(\tilde{e}_2^*)$. The set of pooling equilibria,

as illustrated in figure 8, is composed of all the effort levels comprised in the interval $[\underline{e}, e_2^{*d}]$, where e_2^{*d} is the effort level that makes the incentive constraint hold with equality $\pi(e_2^{*d}) = \pi(\tilde{e}_2^*)$. In figure 8, e_2^{*d} is the value on the x-Axis corresponding to the crossing of *Equilibrium Profits* line with the *Maximum Deviation Profits*. Hence, the set of pooling equilibrium is given by:

$$S_p = \{(e_2^{*g}, e_2^{*b}) | e_2^{*g} = e_2^{*b} = e_2^*, \text{with } e_2^* \in [\underline{e}, e_2^{*d}],$$
$$\text{where } e_2^{*d} \text{ is the solution to } \pi(e_2^{*d}) = \pi(\tilde{e}_2^*), \tilde{e}_2^* = c'^{-1}(bH(N))\}$$

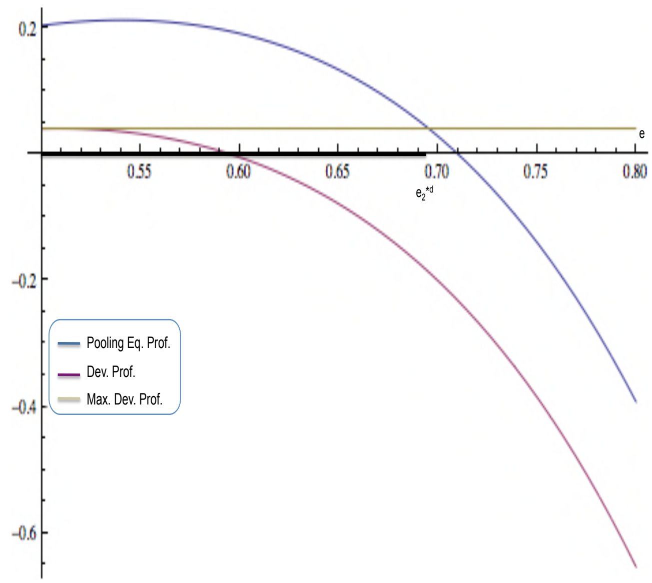

Figure 8: One Period Pooling Equilibrium

Solving for e_2^{*d} depends on the functional form of the cost function. For our choice of cost function, this amounts to solving a third degree equation.

Pooling Equilibrium in the repeated game

We now go back to the first period and analyze the possibility of having a pooling equilibrium, keeping in mind the effect of announced efforts on the updating of beliefs and on the induced probability distribution over future outcomes.

The question we address at this point is, whether, given the set of pooling equilibrium of the second period, can pooling be sustained in the first period as well? The analysis follows a similar pattern as the one above.

Suppose there exist a pooling equilibrium, in which both seller types announce $e_1^g = e_1^b = e_1^*$ at the beginning of the first period, given . As in the last stage, the off-equilibrium-path beliefs assign a zero probability to the seller being of good type,when deviating from e_1^*. Remark that, a deviation in the first period will attribute a "bad seller" label to the deviating seller both in the first *and* in the second period.

For simplicity, assume that the equilibrium effort chosen in the second period is the same, no matter the outcome of the first period. The equilibrium inter-temporal profit of a seller of type θ, in which the efforts are pooled in both periods can be written as:

$$\begin{aligned}
\Pi(e_1^*, e_2^*, \theta) =& \pi_1^* + \delta(\theta e_1^* \pi_2^{*S} + (1-\theta e_1^*)\pi_2^{*F}) \text{ where} \\
\pi_1^* &= (\mu g + (1-\mu)b)e_1^* H(N) \\
\pi_2^{*S} &= (\mu^S g + (1-\mu^S)b)e_2^* H(N) \\
\pi_2^{*F} &= (\mu^F g + (1-\mu^F)b)e_2^* H(N)
\end{aligned}$$

On the equilibrium path, beliefs update according to Bayes rule.

$$\begin{aligned}
\mu^S &= \frac{\mu g}{\mu g + (1-\mu)b} \\
\mu^F &= \frac{\mu(1-ge_1^*)}{\mu(1-ge_1^*) + (1-\mu)(1-be_1^*)}
\end{aligned}$$

When the seller deviates by announcing $\tilde{e_1^*}$ in the first period, the buyers will infer that he is of bad type. This information will be transmitted to the next generation of buyers. Hence, after deviating in the first period, the best choice in the second period is to play $\tilde{e_2^*}$ in the second period. Hence, the inter temporal profits from deviating in the first period are:

$$\Pi^d(\tilde{e^*}_1, \tilde{e^*}_2, \theta) = \pi_1^{*d} + \delta\pi_2^{*d} \text{ where}$$
$$\pi_t^{*d} = bH(N)\tilde{e^*}_t, \; t = 1, 2$$

The maximum attainable profit at deviation, is obtained when employing an effort level $\tilde{e^*}_t = c'^{-1}(bH(N))$. This effort is optimal in both periods, as beliefs don't play any role in this case (for these specified out-of-eq beliefs).

$$\begin{aligned}\Pi^d(\tilde{e^*}_1, \tilde{e^*}_2, \theta) &= (1+\delta)\pi^{*d} \\ &= (1+\delta)(bH(N)c'^{-1}(bH(N)))\end{aligned}$$

We turn back now to characterize the set of pooling equilibrium of the first period.

First, observe that inter temporal profits for the bad type are always lower than those of the good type, i.e. $\min(\Pi(e_1^*, e_2^*, g), \Pi(e_1^*, e_2^*, b)) = \Pi(e_1^*, e_2^*, b)$.

To determine the set of pooling equilibria, we must make sure the announced choice is incentive compatible for both types. Given the above relation between types' profits, the only constraint that needs to hold is:
$\Pi(e_1^*, e_2^*, b) \geq \Pi^d(\tilde{e^*}_1, \tilde{e^*}_2, \theta)$

In the first panel of figure 9, we observe how the intertemporal profits of the bad type (blue area) relate to a deviating seller's profits (green area). For values of e_1^* and e_2^* for which the blue surface is above the green surface, we have that deviation from the pooling equilibrium never pays. The lower figure provides a better picture of the possible pairs of effort levels that can be sustained in a pooling equilibrium. A formal description of this set is given by:

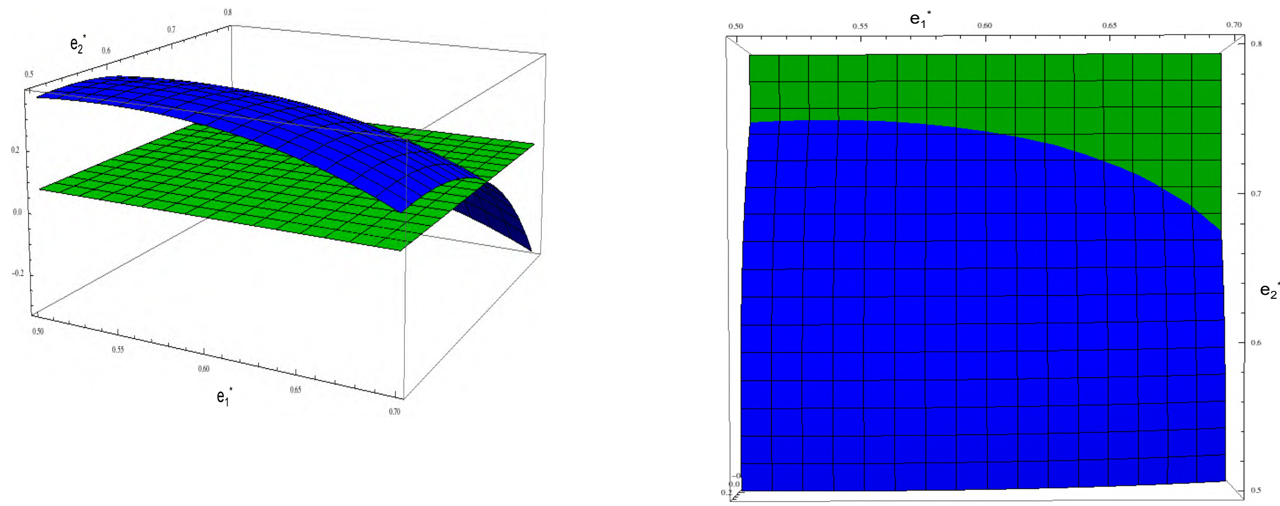

Figure 9: Pooling Eq in the intertemporal game

$$S_{pI} = \{((e_1^{*g}, e_1^{*b}), (e_2^{*g}, e_2^{*b}))|e_2^{*g} = e_2^{*b} = e_2^* \wedge e_1^{*g} = e_1^{*b} = e_1^* \text{such that}$$
$$(e_1^*, e_2^*) \in [\underline{e}, e_2^{*d}] \times [\underline{e}, e_1^{*d}] \text{where } e_2^{*d} \text{and} e_2^{*d}$$
$$\text{are jointly determined from the relation } \Pi(e_1^{*d}, e_2^{*d}) = \pi(\tilde{e}_1^*, \tilde{e}_2^*), \}$$

The points (e_1^{*d}, e_2^{*d}) describe precisely the border between the blue and the green surface in the right panel of figure 9.

D.2 Proposition 1

An equilibrium of the two-periods game where a monopolist of unobservable type chooses higher than minimum effort exists and is unique.

D.3 Proof of Proposition 1

We start by proving that a pooling equilibrium cannot exist.

First, we notice that both types exerting minimal effort level cannot be a solution since the marginal cost of effort is lower than the marginal benefit at this level. Furthermore, given the monotonicity and strict convexity of the cost function and given the linear delivery technology, it follows that a given effort level for both types, the good type has a higher marginal benefit compared to the bad type. In equilibrium the good type will employ more effort than the bad type.

The solution to the monopolist's problem must satisfy following FOCs:

$$\begin{cases} c'(e_1^{*g}) = & gf(e_1^{*g}, e_1^{*b}) \\ c'(e_1^{*b}) = & bf(e_1^{*g}, e_1^{*b}) \end{cases} \tag{34}$$

where $f(e^{*g}, e^{*b}) = \delta H_2^m \underline{e} \left(\mu^S(e_1^{*g}, e_1^{*b}) - \mu^F(e_1^{*g}, e_1^{*b})\right)$ and $H_2^m = \frac{N-1}{N+1}$ is the expected second highest valuation in the monopolist's auction. e_1^{*g}, e_1^{*b} are the equilibrium effort levels of the good type resp. the bad type monopolist in period 1.

First, we prove that $f_{e^g}(e^g, e^b) > 0$ and $f_{e^b}(e^g, e^b) < 0$.

Recall that $\mu^S = \frac{\mu g e^g}{\mu g e^g + (1-\mu) b e^b}$ and $\mu^F = \frac{\mu(1-ge^g)}{\mu(1-ge^g)+(1-\mu)(1-be^b)}$.

It easily follows that $\frac{\partial \mu^S}{\partial e^g} = \frac{\mu(1-\mu)gbe^b}{(\mu ge^g+(1-\mu)be^b)^2} > 0$ and $\frac{\partial \mu^F}{\partial e^g} = \frac{-\mu(1-\mu)gbe^b}{(\mu(1-ge^g)+(1-\mu)(1-be^b))^2} < 0$.
Hence:

$$f_{e^g}(e^g, e^b) = \underbrace{\frac{\partial \mu^S}{\partial e^g}}_{>0} - \underbrace{\frac{\partial \mu^F}{\partial e^g}}_{<0} > 0 \tag{35}$$

Similar analysis applies for $f_{e^b}(e^g, e^b)$: $\frac{\partial \mu^S}{\partial e^b} = \frac{-\mu(1-\mu)gbe^g}{(\mu ge^g+(1-\mu)be^b)^2} < 0$ and
$\frac{\partial \mu^F}{\partial e^b} = \frac{\mu(1-\mu)gbe^g}{\mu(1-ge^g)+(1-\mu)(1-be^b))^2} > 0$.
Hence:

$$f_{e^b}(e^g, e^b) = \underbrace{\frac{\partial \mu^S}{\partial e^b}}_{<0} - \underbrace{\frac{\partial \mu^F}{\partial e^b}}_{>0} < 0 \tag{36}$$

We are looking for an interior solution. Since $c'(\underline{e}) = 0$ and $f(e^g, \underline{e}) > 0$ and resp. $f(\underline{e}, e^b) > 0$, it follows that $c'(\underline{e}) - gf(\underline{e}, e^b) < 0$ and $c'(\underline{e}) - bf(e^g, \underline{e}) < 0$ for $e^g < e^{*g}$and $e^b < e^{*b}$ respectively. In equilibrium 34 holds, namely $c'(e^{*b}) - bf(e^{*g}, e^{*b}) = 0$ and $c'(e^{*g}) - gf(e^{*g}, e^{*b}) = 0$.

It follows that on the $[\underline{e}, e^{*b}] \times [\underline{e}, e^{*g}]$: $c'(e^b) - bf(e^g, e^b)$ is increasing in e^b and $c'(e^g) - gf(e^g, e^b)$ is increasing in e^g.

$$\frac{\partial \left(c'(e^g) - gf(e^g, e^b)\right)}{\partial e^g} |_{e^{*g}} > 0$$
$$\frac{\partial \left(c'(e^b) - bf(e^g, e^b)\right)}{\partial e^b} |_{e^{*b}} > 0 \tag{37}$$

Hence:

$$c''(e^{*g}) > gf_{e^g}(e^{*g}, e^b)$$
$$c''(e^{*b}) > bf_{e^b}(e^g, e^{*b}) \tag{38}$$

$\forall\ e^g \in [\underline{e}, e^{*g}],\ e^b \in [\underline{e}, e^{*b}]$.

We take implicit derivatives of the FOCs in 34

$$\begin{cases} c''(e^{*g})\frac{\partial e^{*g}}{\partial e^b} = & gf_{e^{*g}}(e^{*g}, e^b)\frac{\partial e^{*g}}{\partial e^b}\Big|_{(e^{*g}, e^{*b})} + gf_{e^b}(e^{*g}, e^{*b}) \\ c''(e^{*b})\frac{\partial e^{*b}}{\partial e^g} = & bf_{e^{*g}}(e^g, e^{*b}) + bf_{e^b}(e^{*g}, e^{*b})\frac{\partial e^{*b}}{\partial e^g}\Big|_{(e^{*g}, e^{*b})} \end{cases} \Rightarrow \tag{39}$$

$$\Rightarrow \begin{cases} \left(c''(e^{*g}) - gf_{e^g}(e^{*g}, e^{*b})\right)\frac{\partial e^{*g}(e^b)}{\partial e^b}\Big|_{(e^{*g}, e^{*b})} = & gf_{e^b}(e^{*g}, e^{*b}) > 0 \\ \left(c''(e^{*b}) - bf_{e^b}(e^{*g}, e^{*b})\right)\frac{\partial e^{*b}(e^g)}{\partial e^g}\Big|_{(e^{*g}, e^{*b})} = & bf_{e^g}(e^{*g}, e^{*b}) < 0 \end{cases} \tag{40}$$

From 35, 36 and 38 it follows that $\frac{\partial e^{*g}}{\partial e^b}\Big|_{(e^{*g}, e^{*b})} < 0$ and $\frac{\partial e^{*b}}{\partial e^g}\Big|_{(e^{*g}, e^{*b})} > 0$. Hence best responses are monotonically decreasing respectively increasing. Hence, there exists just one intersection of the best responses, which implies that the equilibrium is unique. ■

D.4 Proposition 2

For any type θ, $\theta \in \{b, g\}$, in the equilibrium with hidden effort and unobserved type, the equilibrium effort level:

1. increases with the discount factor δ

2. The effort level of the good type is increasing in g and decreasing in b. The opposite is true for the effort level of the bad type.

3. is a concave in initial beliefs μ_0

4. For any $g, b, \delta \in [0,1]$ it holds that $e_1^{\theta obsNC} < e_1^{*\theta} \leq e_1^{\theta obsC} < e_1^{\theta eff}$. Moreover,

D.5 Proof of Proposition 2

The first order conditions are given by:

$$c'(e_1^{*\theta}) = \delta\theta H_2^m(\mu_1^S - \mu_1^F)(g-b)\underline{e}$$

1. *For any type, the effort level is increasing in the discount factor δ*

   We take the derivative of the F.O.C. with respect to δ keeping in mind that the equilibrium effort $e_1^{*\theta}$ depends on δ as well:

$$\underbrace{c''(e_1^{*\theta})}_{>0}\frac{\partial e_1^{*\theta}}{\partial \delta} = \underbrace{\theta H_2^m(\mu_1^S - \mu_1^F)(g-b)\underline{e}}_{>0}$$
$$\Leftrightarrow \frac{\partial e_1^{*\theta}}{\partial \delta} > 0$$

2. *For any type, the effort level of the good type is increasing in g and decreasing in b. The opposite is true for the effort level of the bad type.*

   We look first at the first derivatives of μ_1^S and μ_1^F w.r.t g and b.

$$\frac{\partial \mu_1^S}{\partial g} = \frac{\mu b e^b e^g(1-\mu)}{be^b(1-\mu)+e^g g\mu)^2} > 0$$
$$\frac{\partial \mu_1^S}{\partial b} = -\frac{\mu g e^b e^g(1-\mu)}{be^b(1-\mu)+e^g g\mu)^2} < 0$$
$$\frac{\partial \mu_1^F}{\partial g} = -\frac{\mu(1-be^b)e^b e^g(1-\mu)}{(1-(be^b(1-\mu)+e^g g\mu))^2} < 0$$
$$\frac{\partial \mu_1^F}{\partial b} = \frac{\mu(1-ge^g)e^b(1-\mu)}{(1-(be^b(1-\mu)+e^g g\mu))^2} > 0$$

It easily follows that:

$$\underbrace{c''(e_1^{*\theta})}_{>0}\frac{\partial e_1^{*\theta}}{\partial g} = \delta H_2^1(\mu_1^S - \mu_1^F)e^{*\theta obs} + \delta g H_2^1 e^{*\theta obs}\underbrace{\overbrace{\underbrace{\frac{\partial \mu_1^S}{\partial g}}_{>0} - \underbrace{\frac{\partial \mu_1^S}{\partial b}}_{<0}}^{}}_{>0}$$

$$\Leftrightarrow \frac{\partial e_1^{*\theta}}{\partial g} > 0$$

and

$$\underbrace{c''(e_1^{*\theta})}_{>0}\frac{\partial e_1^{*\theta}}{\partial b} = \delta H_2^1(\mu_1^S - \mu_1^F)e^{*\theta obs} + \delta g H_2^1 e^{*\theta obs}\underbrace{\underbrace{\frac{\partial \mu_1^F}{\partial b}}_{<0} - \underbrace{\frac{\partial \mu_1^F}{\partial b}}_{>0}}_{<0}$$

$$\Leftrightarrow \frac{\partial e_1^{*\theta}}{\partial b} < 0$$

3. *For any type, the effort level is first increasing then decreasing in μ_0 (effort level is a concave function of μ_0)*

   (a) We notice first that when $\mu_0 = 0$ or $\mu_0 = 1 \Rightarrow \mu_1^S = \mu_1^F = 0$ respectively $\mu_1^S = \mu_1^F = 1$. This implies in both cases that the RHS of the F.O.Cs is 0. $\Rightarrow c'(e_1^{*\theta}) = 0$. Hence, the optimal effort level when initial beliefs are either 0 or 1 is the maximal costless effort, i.e. $e_1^{*\theta} = e^{*\theta obs}$

   (b) For $\mu_0 > 0$, derive the F.O.Cs implicitly w.r.t. μ_0.

$$\underbrace{c''(e_1^{*\theta})}_{>0}\frac{\partial e_1^{*\theta}}{\partial \mu_0} = \delta\theta H_2^1\frac{\partial(\mu_1^S - \mu_1^F)}{\partial \mu_0}e_2^{*\theta obs}$$

$$\frac{\partial(\mu_1^S-\mu_1^F)}{\partial\mu_0}=\frac{ge^{*g}be^{*b}}{(\mu_0 ge^{*g}+(1-\mu_0)be^{*b})^2}-\frac{(1-ge^{*g})(1-be^{*b})}{(1-(\mu_0 ge^{*g}+(1-\mu_0)be^{*b}))^2} \tag{41}$$

$$\frac{\partial(\mu_1^S-\mu_1^F)}{\partial\mu_0}|_{\mu_0=0}=\underbrace{\frac{ge^{*g}}{be^{*b}}}_{>1}-\underbrace{\frac{(1-ge^{*g})}{(1-be^{*b})}}_{<1}>0 \quad \text{and}$$
$$\frac{\partial(\mu_1^S-\mu_1^F)}{\partial\mu_0}|_{\mu_0=1}=\underbrace{\frac{be^{*b}}{ge^{*g}}}_{<1}-\underbrace{\frac{(1-be^{*b})}{(1-ge^{*g})}}_{>1} \quad \text{and} \tag{42}$$

In the first expression 42 we state the derivative of the RHS of the F.O.Cs w.r.t. μ_0. The denominator of the first fraction is strictly increasing in μ_0, which implies that the first fraction is decreasing in μ_0. The opposite is true for the second fraction. It follows immediately that $\frac{\partial(\mu_1^S-\mu_1^F)}{\partial\mu_0}$ 42 is strictly decreasing in μ_0. It must be then the case that up to a threshold value of μ_0, the RHS of the F.O.Cs is positive, which implies that $\frac{\partial e_1^{*\theta}}{\partial\mu_0}>0$. For values of μ_0 which are higher than the threshold, the RHS of the F.O.Cs turns negative, which implies $\frac{\partial e_1^{*\theta}}{\partial\mu_0}<0$. Hence the optimal effort level is concave in μ_0.

D.6 Proof of Proposition 3

Numerical.

Lemma 4

Single-Crossing condition: Let $e_{j,t}:\Theta_j\times\mathcal{H}^t\longrightarrow[0,1]$ be a decision rule for seller $j=1,2$, as a function of her type $\theta_j\in\Theta_j$ and a history $h^t\in\mathcal{H}^t$. Then, given the multiplicative technology $P(y_j=S|\theta,e)=\theta e$, it is the case in equilibrium that $e_j(g;h^t)\equiv e_j^g(h^t)\geq e_j^b(h^t)\,,\ \forall h^t\in\mathcal{H}^t$, $\forall j$.

D.7 Proof of Lema 4

Define $e^g \in [0,1]$ and $e^b \in [0,1]$ as the equilibrium effort decisions for a seller of type g and type b respectively. Relying on a revealed preference argument, it must then be the case that, for seller j of type θ

$$\Pi_{j,0}\left(e^{\theta}_{j,1};\theta\right) \geq \Pi_{j,0}\left(e^{\theta'}_{j,1,};\theta\right)$$

This implies, for type g and b,

$$c\left(e^g\right) - c\left(e^b\right) \leq \delta g\left(e^g - e^b\right)\left[r_{-j,1}\left(\pi^{SS}_{j,2} - \pi^{FS}_{j,2}\right) + \left(1 - r_{-j,1}\right)\left(\pi^{SF}_{j,2} - \pi^{FF}_{j,2}\right)\right] \quad (43)$$

and

$$c\left(e^g\right) - c\left(e^b\right) \geq \delta b\left(e^g - e^b\right)\left[r_{-j,1}\left(\pi^{SS}_{j,2} - \pi^{FS}_{j,2}\right) + \left(1 - r_{-j,1}\right)\left(\pi^{SF}_{j,2} - \pi^{FF}_{j,2}\right)\right] \quad (44)$$

Here,

$$\pi^{y_{1,1},y_{2,1}}_{j,2} = \underline{\mathrm{e}}\left(g\mu^{y_{j,1}}_{j,1} + b\left(1 - \mu^{y_{j,1}}_{j,1}\right)\right) H_{2,j}\left(r_{1,2}, r_{2,2}\right) - c\left(\underline{\mathrm{e}}\right)$$

is the profit for seller j in period 2 after the outcomes $(y_{j,1}, y_{-j,1})$ have been realized in period 1, and

$$r_{-j,1} = ge^g_{-j,1}\mu_{-j,0} + be^b_{-j,1}\left(1 - \mu_{-j,0}\right)$$

is the *reputation* of seller $-j$ in period 1, with initial type-beliefs given by $(\mu_{j,0}, \mu_{-j,0})$.

Summing eq.(43) and eq.(44) we have

$$\delta\left(e^g - e^b\right)(g - b) \geq 0$$

which, given the assumption $0 < b < g \leq 1$, implies that $e^g \geq e^b > \underline{\mathrm{e}}$. ■

D.8 Proof of Proposition 4

Existence

The equilibrium is described by the following system of equations:

$$\begin{cases} c'\left(e^{*g}_{j,1}\right) & = g\delta\left[r_{-j,1}\left(\pi^{SS}_{j,2}-\pi^{FS}_{j,2}\right)+\left(1-r_{-j,1}\right)\left(\pi^{SF}_{j,2}-\pi^{FF}_{j,2}\right)\right] \\ c'\left(e^{*b}_{j,1}\right) & = b\delta\left[r_{-j,1}\left(\pi^{SS}_{j,2}-\pi^{FS}_{j,2}\right)+\left(1-r_{-j,1}\right)\left(\pi^{SF}_{j,2}-\pi^{FF}_{j,2}\right)\right] \\ c'\left(e^{*g}_{-j,1}\right) & = g\delta\left[r_{j,1}\left(\pi^{SS}_{-j,2}-\pi^{FS}_{-j,2}\right)+\left(1-r_{j,1}\right)\left(\pi^{SF}_{-j,2}-\pi^{FF}_{-j,2}\right)\right] \\ c'\left(e^{*b}_{-j,1}\right) & = b\delta\left[r_{j,1}\left(\pi^{SS}_{-j,2}-\pi^{FS}_{-j,2}\right)+\left(1-r_{j,1}\right)\left(\pi^{SF}_{-j,2}-\pi^{FF}_{-j,2}\right)\right] \end{cases} \tag{45}$$

where $r_{j,1}$ is the perceived probability of success of seller j at time 1, $\pi^{y_j,y_{-j}}_{j,2}$ is the time 2 profit of seller j given that the two players have registered the outcomes $y_j, y_{-j} \in \{S, F\}$

To prove the existence of a MPE of the two-stage game with two sellers we use Brouwer's fixed point theorem.

We start first by characterizing the marginal cost function $c'(\cdot)$ and its inverse $c'^{-1}(\cdot)$.

$c' : [\underline{e}, 1) \to [0, \infty)$ is a monotonically increasing and strictly convex function. Hence $c'(\cdot)$ is a bijection. This guarantees the existence of well defined inverse function $c'^{-1} : [0, \infty) \to [\underline{e}, 1)$.

The above system of FOC's hence translates into:

$$\begin{cases} e^{*g}_{j,1} & = c'^{-1}\left(g\delta\left[r_{-j,1}\left(\pi^{SS}_{j,2}-\pi^{FS}_{j,2}\right)+\left(1-r_{-j,1}\right)\left(\pi^{SF}_{j,2}-\pi^{FF}_{j,2}\right)\right]\right) \\ e^{*b}_{j,1} & = c'^{-1}\left(b\delta\left[r_{-j,1}\left(\pi^{SS}_{j,2}-\pi^{FS}_{j,2}\right)+\left(1-r_{-j,1}\right)\left(\pi^{SF}_{j,2}-\pi^{FF}_{j,2}\right)\right]\right) \\ e^{*g}_{-j,1} & = c'^{-1}\left(g\delta\left[r_{j,1}\left(\pi^{SS}_{-j,2}-\pi^{FS}_{-j,2}\right)+\left(1-r_{j,1}\right)\left(\pi^{SF}_{-j,2}-\pi^{FF}_{-j,2}\right)\right]\right) \\ e^{*b}_{-j,1} & = c'^{-1}\left(b\delta\left[r_{j,1}\left(\pi^{SS}_{-j,2}-\pi^{FS}_{-j,2}\right)+\left(1-r_{j,1}\right)\left(\pi^{SF}_{-j,2}-\pi^{FF}_{-j,2}\right)\right]\right) \end{cases} \tag{46}$$

Denote now $\boldsymbol{e} \equiv \left(e^g_j, e^b_j, e^g_{-j}, e^b_{-j}\right)$ the 4-dimensional vector of efforts (actions) for seller-types, with $\boldsymbol{e} \in \mathcal{E}^4 = [\underline{e}, \bar{e}]^4$. Note that $\mathcal{E}$ is a convex and compact subset of the one-dimensional Euclidean space.

For $\boldsymbol{e} > 0$, the type-belief μ, the success probability function $r(\mu_0, e) \equiv P(y_j = S)$ and the expected valuation of the second highest type in seller's j auction, $H_{j,2}(\boldsymbol{r})$, are all vector-valued functions *continuous* in $\boldsymbol{e}$.

Define the function (for a fixed δ, θ and μ_0) $S_j^\theta : \mathcal{E}^4 \longrightarrow D$ with $D \subset \mathbb{R}$ and $i = 1, 2$ as

$$S_j^\theta(\boldsymbol{e}; \theta, \delta, \mu_0) = \theta\delta\left[r_{-j,1}\left(\pi_{j,2}^{SS} - \pi_{j,2}^{FS}\right) + (1 - r_{-j,1})\left(\pi_{j,2}^{SF} - \pi_{j,2}^{FF}\right)\right]$$

where $\pi_{j,2}^{y_j,y_{-j}} = r_j^{y_j} H_{j,2}$ is the profit for seller j in period 2 after the (joint) outcome for seller j and $-j$ $\boldsymbol{y} = (y_j, y_{-j})$. One can readily see that S_j^θ is a continuous and bounded function.

Hence, also the function composition $\left(c'^{-1} \circ S_j^\theta\right) : \mathcal{E}^4 \longrightarrow \mathcal{E}$ is continuous and bounded. Denote the vector function composition as $(c'^{-1} \circ \boldsymbol{S}) : \mathcal{E}^4 \longrightarrow \mathcal{E}^4$.

So far we have proven that $\mathcal{E}$ is a convex and compact subset of the 4-dimensional euclidean space on which a continuous and bounded function $(c'^{-1} \circ \boldsymbol{S})$ is defined.

Then, Brouwer's fixed point theorem guarantees the existence of a fix point: $\exists \boldsymbol{e}^*$ s.t. $\boldsymbol{e}^* = (c'^{-1} \circ \boldsymbol{S})(\boldsymbol{e}^*)$.

Uniqueness

To prove uniqueness of this equilibrium, we use the results developed by Mason and Valentinyi Mason and Valentinyi (2007). Their paper provides sufficient conditions for existence and uniqueness of equilibrium in monotone pure strategies in a broad class of Bayesian games. When incremental interim payoffs satisfy the *uniform single crossing* and the *Lipschitz continuity* (with respect to other's strategy) condition, then a contraction mapping argument can be applied to prove the existence and uniqueness of the equilibrium.

In our setup, the interim payoff of a seller j (when seller j knows its type θ_j, $\theta_j \in \{g, b\}$) is given by its expected inter-temporal profit.

$$\Pi_j = r_{j,1}H_{2,j}(r_{j,1}, r_{-j,1}) - c(e_{j,1}^{\theta_j}) + \delta[\theta_j e_{j,1}^{\theta_j} r_{-j,1}(r_{j,1}^S H_{2,j}(r_{j,2}^S, r_{-j,2}^S)) + \theta e_{j,1}^{\theta_j}(1-r_{-j,1})(r_{j,1}^S H_{2,j}(r_{j,2}^S, r_{-j,2}^F)))$$
$$(1-\theta_j e_{j,1}^{\theta_j}) r_{-j,1}(r_{j,1}^F H_{2,j}(r_{j,2}^F, r_{-j,2}^S)) + (1-\theta_j e_{j,1}^{\theta_j})(1-r_{-j,1})(r_{j,1}^F H_{2,j}(r_{j,2}^F, r_{-j,2}^F))] \tag{47}$$

Bearing in mind that $r_{j,1} = \mu_j g\tilde{e}_{j,1}^g + (1-\mu_j)b\tilde{e}_{j,1}^b$, and denoting period 1 payoff by $\boldsymbol{R_{j,1}} = r_{j,1}H_{2,j}(r_{j,1}, r_{-j,1})$ and period 2 payoffs respectively $\boldsymbol{R_{j,2}}(y_j, y_{-j}) = r_{j,2}^{y_j} H_{2,j}(r_{j,2}^{y_j}, r_{-j,2}^{y_{-j}})$ where $y_j, y_{-j} \in \{S, F\}$ represent the outcomes of the transactions between period 1 and period 2, we can rewrite the expression in the following way:

$$\begin{aligned}\Pi_j = \mu_{-j,1}\Bigg[&\boldsymbol{R_{j,1}} - c(e_{j,1}^{\theta_j}) + \delta\Bigg[g\tilde{e}_{-j,1}^g\Bigg(\theta_j e_{j,1}^{\theta_j}\boldsymbol{R_{j,2}}(S,S) + (1-\theta_j e_{j,1}^{\theta_j})\boldsymbol{R_{j,2}}(F,S)\Bigg) + \\ &+ (1-g\tilde{e}_{-j}^g)\Bigg(\theta_j e_{j,1}^{\theta_j}\boldsymbol{R_{j,2}}(S,F) + (1-\theta_i e_{j,1}^{\theta_j})\boldsymbol{R_{j,2}}(F,F)\Bigg)\Bigg]\Bigg] + \\ (1-\mu_{-j,1})\Bigg[&\boldsymbol{R_{j,1}} - c(e_{j,1}^{\theta_j}) + \delta\Bigg[b\tilde{e}_{-j}^b\Bigg(\theta_j e_{j,1}^{\theta_j}\boldsymbol{R_{j,2}}(S,S) + (1-\theta_j e_{j,1}^{\theta_j})\boldsymbol{R_{j,2}}(F,S)\Bigg) + \\ &+ (1-b\tilde{e}_{-j}^b)\Bigg(\theta_j e_{j,1}^{\theta_j}\boldsymbol{R_{j,t}}(S,F) + (1-\theta_j e_{j,1}^{\theta_j})\boldsymbol{R_{j,2}}(F,F)\Bigg)\Bigg]\Bigg]\end{aligned} \tag{48}$$

Further, we define the *ex-post* payoff function π :

$$\begin{aligned}\pi_j(e_{j,1}^{\theta_j}, \theta_j, \tilde{e}_{-j,1}^{\theta_{-j}}, g) = &\boldsymbol{R_{j,1}} - c(e_{j,1}^{\theta_j}) + \delta\Bigg[g\tilde{e}_{-j}^g\Bigg(\theta_j e_{j,1}^{\theta_j}\boldsymbol{R_{j,t}}(S,S) + (1-\theta_j e_{j,1}^{\theta_j})\boldsymbol{R_{j,2}}(F,S)\Bigg) + \\ &+ (1-g\tilde{e}_{-j}^g)\Bigg(\theta_j e_{j,1}^{\theta_j}\boldsymbol{R_{j,2}}(S,F) + (1-\theta_j e_{j,1}^{\theta_j})\boldsymbol{R_{j,2}}(F,F)\Bigg)\Bigg]\end{aligned} \tag{49}$$

and hence the expression 48 can be rewritten as:

$$\Pi_j(e_{j,1}^{\theta_j}, \theta_{-j}, \tilde{e}_{-j,1}^{\theta_{-j}}, \theta_{-j}) = \sum_{\theta_{-j}\in\{g,b\}} \pi_j(e_{j,1}^{\theta_j}, \theta_j, \tilde{e}_{-j,1}^g, \theta_{-j}) f(\theta_{-j}|\theta_j) \tag{50}$$

where $f(\theta_{-j}|\theta_j)$ is the conditional density of j's types.

Further, we define the incremental *ex-post* payoffs by:

$$\Delta\pi_j(e_j^{\theta_j}, e_j^{'\theta_j}, \theta_j; \tilde{e}_{-j}^{\theta_{-j}}, \theta_{-j}) \equiv \pi_j(e_j^{\theta_j}, \theta_j; \tilde{e}_{-j}^{\theta_{-j}}, \theta_{-j}) - \pi_j(e_j^{'\theta_j}, \theta_j; \tilde{e}_{-j}^{\theta_{-j}}, \theta_{-j}) \tag{51}$$

Given the assumptions on the cost function, the *ex-post* payoff function satisfies the following property as in Mason and Valentinyi (2007)

P1 The payoff function $\pi_j : [0,1] \times \{g,b\} \to \mathbf{R}$ is bounded and measurable, and (upper semi-) continuous in own action. The types have conditional densities w.r.t the Lebesgue measure. The conditional density of θ_{-j} given θ_j , $f(\theta_{-j}|\theta_j)$ is strictly positive.

Given this property, we proceed to present conditions U1-U3 and D1-D2 as in Mason and Valentinyi (2007). These conditions will enable us to prove uniqueness of the equilibrium.

U1 Uniformly Positive Sensitivity to Own Action and Type. There is a $\gamma \in (0,\infty)$ s.t. **for all** $e^{\theta_j}_{j,1} \geq e^{'\theta_j}_{j,1}$,$\theta_j \geq \theta'_j$, $\tilde{e}^{\theta_{-j}}_{-j}$,$\theta_{-j}$

$$\underbrace{\Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \theta_j; e^{\theta_{-j}}_{-j,1}, \theta_{-j}) - \Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \theta'_j; \tilde{e}^{\theta_{-j}}_{-j}, \theta_{-j})}_{\equiv \Delta_{\theta_j}\Delta u\left(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \tilde{e}_{-j,1}, \theta_j, \theta_{-j}\right)} \geq \gamma(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1}) d_T(\theta_j, \theta'_j). \tag{52}$$

We assume $d_T(\theta_j, \theta'_j) = |\theta_j - \theta'_j|$

Following the notation above and expression 49, the ex-post incremental payoff for seller j (IP_j) for $e^{\theta_j}_j > e^{'\theta_j}_j$ is given by

$$\begin{aligned}
&\Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \theta_j; e^{\theta_{-j}}_{-j,1}, \theta_{-j}) = \\
&\quad = -\left(c\left(e^{\theta_j}_{j,1}\right) - c\left(e^{'\theta_j}_{j,1}\right)\right) + \delta\Bigg(E\left[\boldsymbol{R_{j,2}}\left(y_j, y_{-j}\right)\Big| e^{\theta_j}_{j,1}, \tilde{e}^{\theta_{-j}}_{-j,1}, \boldsymbol{\theta}\right] - E\left[\boldsymbol{R_{j,2}}\left(y_j, y_{-j}\right)| e^{'\theta_j}_{j,1}, \tilde{e}^{\theta_{-j}}_{-j,1}, \boldsymbol{\theta}\right.\\
&\quad = -\left(c\left(e^{\theta_j}_{j,1}\right) - c\left(e^{'\theta_j}_{j,1}\right)\right) + \delta\theta_j(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1})\Bigg(\theta_{-j}\tilde{e}^{\theta_{-j}}_{-j,1}\Big(\boldsymbol{R_{j,2}}\left(S,S\right) - \boldsymbol{R_{j,2}}\left(F,S\right)\Big) + \\
&\qquad\qquad + \left(1 - \theta_{-j}\tilde{e}^{\theta_j}_{-j,1}\right)\Big(\boldsymbol{R_{j,2}}\left(S,F\right) - \boldsymbol{R_{j,2}}\left(F,F\right)\Big)\Bigg)
\end{aligned}$$

It follows then that

$$\begin{aligned}
\Delta_{\theta_j}\Delta u\left(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \tilde{e}^{\theta_{-j}}_{-j,1}, \theta_j, \theta_{-j}\right) =& \delta(\theta_j - \theta'_j)(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1})\Bigg(\theta_{-j}\tilde{e}^{\theta_{-j}}_{-j,1}\Big(\boldsymbol{R_{j,2}}\left(S,S\right) - \boldsymbol{R_{j,2}}\left(F,S\right)\Big) + \\
&+ \left(1 - \theta_{-j}\tilde{e}^{\theta_j}_{-j,1}\right)\Big(\boldsymbol{R_{j,2}}\left(S,F\right) - \boldsymbol{R_{j,2}}\left(F,F\right)\Big)\Bigg)
\end{aligned}$$

Proving that condition **U1** holds is equivalent to proving that $\exists \ \gamma \in (0,\infty)$ s.t. **for all** $e^{\theta_j}_{j,1} \geq e^{'\theta_j}_{j,1}, \theta_j \geq \theta'_j, \ \tilde{e}^{\theta_{-j}}_{-j}, \ \theta_{-j}$

$$\underbrace{\delta\left(\theta_{-j}\tilde{e}^{\theta_j}_{-j,1}\Big(\boldsymbol{R_{j,2}}\left(S,S\right) - \boldsymbol{R_{j,2}}\left(F,S\right)\Big) + \Big(1-\theta_{-j}\tilde{e}^{\theta_{-j}}_{-j,1}\Big)\Big(\boldsymbol{R_{j,2}}\left(S,F\right) - \boldsymbol{R_{j,2}}\left(F,F\right)\Big)\right)}_{\equiv A} \geq \gamma \tag{53}$$

$$A = \delta\left(\theta_{-j}\tilde{e}^{\theta_{-j}}_{-j,1}\Big(r^S_{j,2}H_{2,j}\left(S,S\right) - r^F_{j,2}H_{2,j}\left(F,S\right)\Big) + \Big(1-\theta_{-j}\tilde{e}^{\theta_{-j}}_{-j,1}\Big)\Big(r^S_{j,2}H_{2,j}\left(S,F\right) - r^F_{j,2}H_{2,j}\left(F,F\right)\Big)\right) \tag{54}$$

The measure for the success probability $r = \mu g\tilde{e}^g + (1-\mu)b\tilde{e}^b$ is bounded above by $\bar{r} = g$ and bounded below by $\underline{\mathrm{r}} = b\underline{\mathrm{e}}$. Since the $H_{2,j}(r_{j,1}, r_{-j,1})$ is monotonically increasing in the first variable and decreasing in the second, we can define the lower bound for the $\underline{\mathrm{H}}_{2,j} = H_{2,j}(\underline{\mathrm{r}}, \bar{r})$. The upper bound would be then $\bar{H}_{2,j} = H_{2,j}(\bar{r}, \underline{\mathrm{r}})$. It is straightforward then that also A is bounded.

In what follows, we will need to determine a lower bound for A given the parameters N, g, b, δ. We call this lower bound γ. In the following Theorem 1 we will numerically determine values for γ together with other boundaries, for given parameter values N, g, b, δ.

U2 Lipschitz Continuity in Own Action There is an $\omega \in (0,\infty)$ s.t. **for all** $e^{\theta_j}_{j,1} \geq e^{'\theta_j}_{j,1}, \theta_j \geq \theta'_j, \ \tilde{e}^{\theta_{-j}}_{-j,1}, \ \theta_{-j,1}$

$$|\Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \tilde{e}^{\theta_{-j}}_{-j,1}, \theta_j, \theta_{-j})| \leq \omega(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1}).$$

The cost function and marginal cost function are monotonically increasing and bounded on the domain $[0,1]$, $\max\{c(1), c'(1)\} < C$. Hence given any two actions $e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1} \in [0,1]$, $e^{\theta_j}_{j,1} > e^{'\theta_j}_{j,1}$ there exists an $e^{\tilde{\theta}_j}_{j,1} \in \{e^{'\theta_j}_{j,1}, \ e^{\theta_j}_{j,1}\}$ such that $c'(e^{\tilde{\theta}_j}_{j,1})(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1}) = c\left(e^{\theta_j}_{j,1}\right) - c\left(e^{'\theta_j}_{j,1}\right)$. On the other hand, we know that $A \leq H_{2,j}(S,F) \leq H_{2,j}(\bar{r}, \underline{\mathrm{r}}) \equiv \bar{H_{2,j}}$. It hence follows that:

$$
\begin{aligned}
|\Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \tilde{e}^{\theta_{-j}}_{-j}, \theta_j, \theta_{-j})| =& | - (c\left(e^{\theta_j}_{j,1}\right) - c\left(e^{'\theta_j}_{j,1}\right)) + \delta\theta_j(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1})A| \\
&\leq | - (c\left(e^{\theta_j}_{j,1}\right) - c\left(e^{'\theta_j}_{j,1}\right))| + |\delta\theta_j(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1})A| \\
&\leq c'(e^{\tilde{\theta}_j}_{j,1})(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1}) + \delta\theta_j\bar{H_{2,j}}(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1}) \\
&= \left(c'(e^{\tilde{\theta}_j}_{j,1}) + \delta\theta_j\bar{H_{2,j}}\right)(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1}) \\
&\leq \left(C + \delta\theta_j\bar{H_{2,j}}\right)(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1})
\end{aligned}
$$

We can now define the parameter $\omega = C + \delta\theta_j\bar{H_{2,j}}$ s.t.

$$
|\Delta\pi_j(e^{\theta_j}_{j,1}, e'\theta_{j_{j,1}}, \tilde{e}^{\theta_{-j}}_{-j}, \theta_j, \theta_{-j})| \leq \omega(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1})
$$

U3 Uniformly Bounded Sensitivity to Opponent's Action. There is a $\kappa \in (0,\infty)$ s.t. **for all** $e^{\theta_j}_{j,1} \geq e^{'\theta_j}_{j,1}$,$\theta_j \geq \theta'_j$, $\tilde{e}^{\theta_{-j}}_{-j,1}$, θ_{-j}

$$
\underbrace{|\Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \theta_j; \tilde{e}^{\theta_{-j}}_{-j,1}, \theta_{-j}) - \Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \theta_j; \tilde{e}^{'\theta_{-j}}_{-j,1}, \theta_{-j})|}_{|\Delta_{\tilde{e}^{\theta_{-j}}_{-j,1}}\Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \theta_j; \tilde{e}^{\theta_{-j}}_{-j,1}, \theta_{-j})|} \leq \kappa(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1})
$$

$$
\begin{aligned}
|\Delta_{\tilde{e}^{\theta_{-j}}_{-j,1}}\Delta\pi_j(e^{\theta_j}_{j,1}, e^{'\theta_j}_{j,1}, \theta_j; \tilde{e}^{\theta_{-j}}_{-j,1}, \theta_{-j})| =& \Bigg|\delta\Bigg(\Big(\boldsymbol{R_{j,2}}(S,S) - \boldsymbol{R_{j,2}}(F,S)\Big) - \\
&-\Big(\boldsymbol{R_{j,2}}(S,F) - \boldsymbol{R_{j,2}}(F,F)\Big)\Bigg)\theta_j\theta_{-j}\left(\tilde{e}^{\theta_{-j}}_{-j,1} - e^{'\theta_{-j}}_{-j,1}\right)\left(e^{\theta_j}_{j,1} - e^{'\theta_j}_{j,1}\right)\Bigg|
\end{aligned}
$$

Similar reasoning as in **U1** applies to conclude that the term

$$
\underbrace{\left|\delta\left(\Big(\boldsymbol{R_{j,2}}(S,S) - \boldsymbol{R_{j,2}}(F,S)\Big) - \Big(\boldsymbol{R_{i,2}}(S,F) - \boldsymbol{R_{i,2}}(F,F)\Big)\right)\theta_i\theta_{-i}\left(\tilde{e}^{\theta_{-j}}_{-j} - \tilde{e}^{'\theta_{-j}}_{-j}\right)\right|}_{\equiv K} \tag{55}
$$

is bounded. We denote the upper bound of this term by κ. In what follows, we will need to determine an upper bound for K given the parameters N, g, b, δ. In Theorem 4 we will numerically determine values for κ together with other boundaries, for given parameter values N, g, b, δ.

D1 There is a $\iota \in (0,\infty)$ s.t. for any $\theta_k > \theta_k^{'}$ and $k \in \{j,-j\}$ $\sqrt{\mathcal{I}(\theta_k,\theta_k^{'})} \leq \iota d_T(\theta_k,\theta_k^{'})$, where

$$\mathcal{I}(\theta_k,\theta_k^{'}) \equiv Var_{T_{-k}}\left(\frac{f(\theta_{-k}|\theta_k) - f(\theta_{-k}|\theta_k^{'})}{f(\theta_{-k}|\theta_k)}\right)$$

This is a condition on the *Fisher's information* of a player's type about the types of the opponents. Note that in our case, we assume *independence* of types. In other words, $f\left(\theta_{-k}\middle|\theta_k = g\right) = f\left(\theta_{-k}\middle|\theta_k = b\right)$ which implies that $\mathcal{I}(\theta_k,\theta_k^{'}) = 0$. Therefore, the condition

$$\sqrt{\mathcal{I}(\theta_k,\theta_k^{'})} \leq \iota\,(g-b)$$

holds for any $\iota \in (0,\infty)$.

D2 There is a $\nu \in [0,\infty)$ s.t. $f_{-j}(\theta_{-j}|\theta_j) \leq \nu$ for all $j \neq -j$. This is a condition on the (ex-ante) heterogeneity of types. We are looking for some $\nu \in [0,\infty)$ such that $f_{-j}\left(\theta_{-j}\middle|\theta_j\right) \leq \nu \ \forall j,-j$. In terms of initial beliefs, this implies for $k \in \{j,-\}$

$$\begin{cases} \mu_k & \leq \nu \\ 1-\mu_k & \leq \nu \end{cases}$$

$\Rightarrow$ Any $\nu \in [1,\infty]$ satisfies this condition.

Theorem 4: *If assumptions U1-U3 and D1-D2 hold, and if*

$$\gamma > \iota\omega + \upsilon\kappa \tag{56}$$

then the best response (BR) correspondence is a contraction, and hence there is a unique equilibrium of the Bayesian game. Furthermore, this equilibrium is in monotone pure strategies.

In compliance with conditions **U1-U3** and **D1-D2**, we now determine value of the parameters $N, g, b,$ and δ (to determine the values for γ and κ) for which the relation 56 holds. We recall that:

$$\omega = C + \delta\theta_j \bar{H_{2,j}}, \text{where } C \leq c'(1)$$
$$\iota = \epsilon$$
$$\nu = 1$$

where we have chosen ϵ to be arbitrarily small, $\epsilon \to 0$.
We investigate now how the maximal value of K (henceforth denoted by $\bar{K} = \max K$)in expression 55 and the minimal value of A in 53 (henceforth denoted by $\underline{\mathrm{A}} = \min A$) vary with the parameters N, g, b and δ. Our target is to pin down restrictions for these parameters for which $\bar{K} < \underline{\mathrm{A}}$

D.9 Proof of Proposition 5

Numerical.

E The Value of Reputation in Infinite Horizon

Estimating the effect of reputation on revenues and sales is a key question for both market participants (sellers) and market makers or regulators (eBay, etc). The latter need to design a way for buyers to have information about the reliability of sellers, while reliable sellers will be reliable depending on how outcomes and information might affect their revenues. Reputation mechanisms are considered to be a solution to the market failures arising from the different sources of inherent asymmetric information. Concretely, and in the spirit of Akerlof (1970), a well-functioning reputation / feedback mechanism might help welfare improving trades that where previously not happening to materialize, and in the process allow "good"/ "truthful" sellers to increase their revenues.

In this section we are interested in quantifying the bias that might arise in the *reduced form* estimation of the impact of reputation on short-run seller revenues when reputation building and strategic behavior are not incorporated into the identification strategy. Although recent empirical studies using both longitudinal (panel) data on sellers and "field" experiments, such as Resnick et al. (2006) and Cabral and Hortaçsu (2010), have gone a long way in trying to identify such effect, we argue that their omission of the above two behaviors might render their estimates misleading.

First, strategic behaviors can generate reputations of competing sellers that are *correlated* across time and a determinant of revenues. Under such scenario, not accounting for competitors' information will bias the estimation of the reputation-elasticity of revenues. The idea is simple: in our setting, when buyers select an auction and bid, it is in their best interest to judge the trustworthiness of *all* sellers through the available public information. The publicly observed reputation of competitors then helps seller j predict the behavior of *future potential* buyers and other seller, behaviors which will determine her revenues. Seller j's effort strategies, therefore, will take into account how current beliefs will affect the updating process of buyers[14], and how this will affect the pool of future buyers that bid in her auction.[15]

Second, Resnick et al. (2006) does not control for the kind of reputation building behavior that has been characterized in seminar works such as Holmstrom (1999).

We will frame the analysis within an infinite horizon version of the game described in the previous sections; this will allow us to incorporate the main mechanisms into an environment where "professional" sellers care about the medium-long term when making decisions. A potential drawback with the benchmark environment is is that it is stripped from many relevant characteristics of on-line markets and industries in general, such as a large number of sellers or the possibility to exit and re-enter the market. Nonetheless, it still allows us to do a first quantification of the endogeneity problem.

In a second step, we will extend the benchmark model by incorporating more

[14] Since seller $-j$ acts in the same way, seller's j best response to her competitors actions and buyers behavior is a best response to current reputation levels. This is simply the main insight from a Markov equilibrium.

[15] Perhaps the strongest assumption here is both the rational expectation of sellers and buyers and their optimal use of all the available information. One could argue that such processing capacity is not realistic, or that the formation of expectations by sellers follows a more reduced form version of the fully rational one. Such considerations, though certainly valid and perhaps even accurate, are outside the scope of this paper.

seller into the market, and maintaining the assumption that both sellers and buyers behave strategically. In order to keep the problem computationally tractable, Our approach will follow the recent developments in computable models of industry dynamics, recently reviewed by Doraszelski and Pakes (2007). In these lines, in the multiple seller setup we assume that each seller only

E.1 Strategic Competition: Recursive Formulation

We first present the infinite horizon version of the problem presented in section 3, casted in terms of reputation (beliefs about seller types) as well as the type itself as state variables.

Belief $\boldsymbol{\mu}_\tau \equiv \{\mu_{j,\tau}, \mu_{-j,\tau}\}$ are a *sufficient statistic* about the sequence of outcomes up to $\tau - 1$. Given that the outcomes of sellers are (conditionally) independent, the recursive formulation of problem for seller j is

$$\begin{aligned} V_j^{comp}\left(\mu_j, \mu_{-j}; \theta\right) = \quad & \max_{e_j^\theta} \Bigg\{ \boldsymbol{R}\left(\mu_j, \mu_{-j}, \boldsymbol{e}^*\right) - c\left(e_j^\theta\right) + \\ & +\delta \sum_{y_j} \sum_{y_{-j}} p\left(y_j, y_{-j} | e_j^\theta, r_{-j}\right) V_j^{comp}\Big(\mu_j', \mu_{-j}'; \theta\Big) \Bigg\} \\ s.t. \quad & e_j^\theta \in [0,1] \end{aligned} \tag{57}$$

where $\boldsymbol{e}^* \equiv \left\{e_j^{g*}, e_j^{b*}, e_{-j}^{g*}, e_{-j}^{b*}\right\}$ and $V_j^{comp}(\cdot)$ is the continuation value for seller j after the outcomes of both sellers, $\{y_j, y_{-j}\}$, which are publicly observable. Recall that, due to Bayesian learning, μ_j' and μ_{-j}' are functions of $\boldsymbol{e}^*, \boldsymbol{\mu}$ and y_j, y_{-j}. In addition, r_{-j} is a function of both μ_{-j} and $\boldsymbol{e}^*_{-j}$. The notation in (57) implies that $\boldsymbol{e}^*$ are effort decisions as perceived by other sellers and buyers, which will be confirmed in an equilibrium: since all players have rational expectations, each one can solve the problem of the other and therefore "guess" an effort choice.

E.1.1 No Reputation and Perfect Reputation as Absorbing States

It is important to note that, due to the assumption about sellers / buyers being Bayesian learners, the Markov process has two *absorbing states*: $\mu = 0$ and $\mu = 1$.

That is,

$$\mu_{j,t} \equiv P\left(\theta_j = g | \mu_{j,t-1} = 0, y_{j,t-1}\right) = 0 \quad \forall\, t, y_{j,t-1}, \mu_{-j,t} \tag{58}$$

and

$$\mu_{j,t} \equiv P\left(\theta_j = g | \mu_{j,t-1} = 1, y_{j,t-1}\right) = 1 \quad \forall\, t, y_{j,t-1}, \mu_{-j,t} \tag{59}$$

Note that expressions (58) and (59) hold for *any* effort level of both sellers. Given the properties of the cost function, the optimal effort under such states is the minimum effort $\underline{e}$. It is then straightforward, given that $c\left(\underline{e}\right) = 0$, to compute the value for a seller at these states:

Proposition 6 *The continuation values at each absorbing state satisfy*

$$V_j^{comp}\left(1, 0; \cdot\right) > V_j^{comp}\left(1, 1; \cdot\right) > V_j^{comp}\left(0, 0; \cdot\right) > V_j^{comp}\left(0, 1; \cdot\right) \tag{60}$$

Proof. Each term in the relation (60) represents the value of selling under competition when the type-beliefs of *both* sellers have converged to either 0 or 1. Such beliefs need not coincide with the *true* type of sellers; indeed, the value at an absorbing state / belief for seller j is *independent* of her true type. Then, we have

$$V_j^{comp}\left(0, 0; \theta_j\right) = \frac{\mathbb{E}\left(b_j^{(2)}\middle|\mu_j = 0, \mu_{-j} = 0\right)}{1-\delta} = \frac{r_j(0,0)\mathbb{E}\left(v_j^{(2)}\middle|\mu_j = 0, \mu_{-j} = 0\right)}{1-\delta} \tag{61}$$

$$V_j^{comp}\left(0, 1; \theta_j\right) = \frac{\mathbb{E}\left(b_j^{(2)}\middle|\mu_j = 0, \mu_{-j} = 1\right)}{1-\delta} = \frac{r_j(0,1)\mathbb{E}\left(v_j^{(2)}\middle|\mu_j = 0, \mu_{-j} = 1\right)}{1-\delta} \tag{62}$$

$$V_j^{comp}\left(1, 0; \theta_j\right) = \frac{\mathbb{E}\left(b_j^{(2)}\middle|\mu_j = 1, \mu_{-j} = 0\right)}{1-\delta} = \frac{r_j(1,0)\mathbb{E}\left(v_j^{(2)}\middle|\mu_j = 1, \mu_{-j} = 0\right)}{1-\delta} \tag{63}$$

$$V_j^{comp}\left(1, 1; \theta_j\right) = \frac{\mathbb{E}\left(b_j^{(2)}\middle|\mu_j = 1, \mu_{-j} = 1\right)}{1-\delta} = \frac{r_j(1,1)\mathbb{E}\left(v_j^{(2)}\middle|\mu_j = 1, \mu_{-j} = 1\right)}{1-\delta} \tag{64}$$

where $r_j(0,0) = r_j(0,1) = \underline{e} * b < r_j(1,0) = r_j(1,1) = \underline{e} * g$ are the perceived probabilities of successfull delivery. Recall that the properties of the function $H_{2j}(\mu_j, \mu_{-j}) \equiv \mathbb{E}\left(v_j^{(2)}\middle|\mu_j, \mu_{-j}\right)$ (the expected 2^{nd} highest valuation in seller j's auction when type beliefs are (μ_j, μ_{-j}) are such that $H_{2j}\left(\mu_j, \mu_{-j}\right)$ is increasing in μ_j, decreasing in μ_{-j} and $H_{2j}(\cdot,\cdot) = H_{2-j}(\cdot,\cdot)$. The result then follows inmediately. ■

Relation (60) says that, as one would expect, the highest value for seller j of selling under competition is realized when the belief other market participants have about her are maximal (i.e. 1) and the beliefs about her competitor are minimal. In such case, she can enjoy the returns to a perfect reputation even by putting minimum effort, without fearing about her competitor. On the other side of the spectrum, her continuation value is minimal when her belief is minimal and her competitor enjoys perfect reputation. For the middle cases, the relation $V_j^{comp}(1,1;\cdot) > V_j^{comp}(0,0;\cdot)$ arises since, although the expected second highest valuation is the same when $(\mu_j, \mu_{-j}) = (0,0)$ and $(\mu_j, \mu_{-j}) = (1,1)$, a higher perceived probability of successful delivery $r_j(\cdot,\cdot)$ incentivizes each buyer to bid higher.

Note that whether relation (60) is enough in order for a bad seller to try to disguise himself as a good seller in any state, will depend on the marginal value of effort at different type beliefs.

E.1.2 Optimal Effort and the Euler Equation

As in the two period case, the effort best response of sellers $j = 1,2$ and types $\theta \in g, b$ are implicitly characterized by the system of FOCs:

$$\begin{aligned} c'\left(e_j^\theta\right) &= \delta\theta\Bigg(r_{-j}\Big(V_j^{comp}\left(\mu_j'(S), \mu_{-j}'(S);\theta\right) - V_j^{comp}\left(\mu_j'(F), \mu_{-j}'(S);\theta\right)\Big) + \\ &\quad + (1-r_{-j})\Big(V_j^{comp}\left(\mu_j'(S), \mu_{-j}'(F);\theta\right) - V_j^{comp}\left(\mu_j'(F), \mu_{-j}'(F);\theta\right)\Big)\Bigg) \qquad (65) \\ &\quad j = 1,2\,;\ \theta = g, b \end{aligned}$$

where $\mu_j'(y)$ is the updated reputation after an outcome y in the current transaction.

From the discussion in the previous subsection, it is clear that whenever $\mu_j = 0$ or $\mu_j = 1$, the right hand side of (66) is 0, for any μ_{-j}. Therefore, in those cases $e_j^\theta = \underline{e}$. In the rest of the state space, the intuition for (66) is relatively straightforward. Given that effort decisions affect directly the probability of a successful outcome, the marginal benefit of effort is given by the difference in continuation value between a success and a failure, for all the possible outcomes of the competitor, weighted by the marginal change in the probability of a success. A seller will be willing to increase her effort as long as the (discounted) incremental difference in the continuation value, weighted by the marginal effect of effort on the outcome probabilities θ, is big enough. Seller j chooses effort strategically taking into account two pieces of information: (i) what buyers perceive about other types of sellers, which will be part of the updating process leading to μ_j', and (ii) how the "intensity" of competition (given by the beliefs about her competitor's type) will affect her expected revenues, which affects both r_{-j} and μ_{-j}' in the RHS of (66). In this setting, the expression for the Euler equation for a seller j of type θ is non-standard. The reason is the following: an effort decision after auction t , e_t, will have a probabilistic effect on the state tomorrow, μ'. Importantly, however, this change in the distribution of μ' will affect future states through two channels: future decisions of effort based on μ' which will shape the distribution of μ'', μ''', etc, and through the updating process, which will mean that the effects of e_t will persist.

In order to save on notation below, lets consider, without loss of generality, the case in which the outcome y can take a continuum of values. When then can write the FOC (66) for seller j and type θ_j as

$$-c'\left(e_j^{\theta}\right)+\delta\int_{y_{-j}}\int_{y_j}V^{comp}\Big(\mu_j',\mu_{-j}';\theta\Big)\frac{\partial f\left(y_j|e_j,\theta\right)}{\partial e_j}\Big|_{e_j=e_j^{\theta}}f\left(y_{-j}|\boldsymbol{e_{-j}^{*}}\right)\mathrm{d}y_j\mathrm{d}y_{-j}=0 \tag{66}$$

where e^{θ} is the optimal effort strategy for a seller of type θ, and $f\left(y|e,\theta\right)$ is the probability density for the outcome y conditional on e and θ.

If we repeatedly replace $V^{comp}\left(\cdot\right)$ in the expression above, we get

$$\begin{aligned} c'\left(e^{\theta}\right) \;=\;& \delta\int_{\boldsymbol{y}}\frac{\partial f\left(y_j|e_j,\theta\right)}{\partial e_j}\Big|_{e_j=e_j^{\theta}}\pi_j\Big(\boldsymbol{\mu'},\boldsymbol{e'}^{*};e'\Big)\mathrm{d}\boldsymbol{y}+ \\ &+\delta^2\int_{\boldsymbol{y}}\frac{\partial f\left(y_j|e_j,\theta\right)}{\partial e_j}\Big|_{e_j=e_j^{\theta}}\int_{\boldsymbol{y'}}\pi_j\Big(\boldsymbol{\mu''},\boldsymbol{e''}^{*};e''\Big)f\left(\boldsymbol{y'}|\boldsymbol{e''},\theta\right)\mathrm{d}\boldsymbol{y'}\mathrm{d}\boldsymbol{y}+ \\ &+... \end{aligned} \tag{67}$$

where $\boldsymbol{y}\equiv\{y_j,y_{-j}\}$, $\boldsymbol{\mu}\equiv\{\mu_j,\mu_{-j}\}$, $\boldsymbol{e}\equiv\{\boldsymbol{e_j},\boldsymbol{e_{-j}}\}$ and $\pi\left(\boldsymbol{\mu},\boldsymbol{e}^{*};e\right)=\boldsymbol{R}\left(\mu,\boldsymbol{e}^{*}\right)-$

$c(e)$ is the stage-payoff. In a Markov Perfect equilibrium, of course, the effort decisions $\boldsymbol{e}$ will be a function of current beliefs $\boldsymbol{\mu}$.

Although in this case the derivation of the Euler eq. does not require to work explicitely with the derivative of V^{comp} with respect to the state μ, each seller *does* care about the slope of the value function when deciding an effort level[16]. Therefore, it is still useful to have a look at the slope of $V^{comp}(\mu_j.\mu_{-j})$. Assuming differentiability of $V^{comp}(\boldsymbol{\mu};\cdot)$, the *envelope condition*, describing how the *maximum* value for a seller changes when the state μ varies, is given by

$$\begin{aligned}\frac{\partial V^c(\boldsymbol{\mu};\theta)}{\partial \mu_j} &= \frac{\partial \boldsymbol{R}\Big(\boldsymbol{\mu}, \boldsymbol{e}(\boldsymbol{\mu})\Big)}{\partial \mu_j} + \\ &+ \frac{\partial e_j^\theta(\boldsymbol{\mu})}{\partial \mu_j}\left(-c'\left(e_j^\theta(\boldsymbol{\mu})\right) + \delta \int_{y_{-j}}\int_{y_j} V^c(\boldsymbol{\mu'};\theta)\frac{\partial f(y_j|e_j,\theta)}{\partial e_j}\Big|_{e_j=e_j^\theta} f\left(y_{-j}|\boldsymbol{e^*_{-j}}\right)\mathrm{d}y_j\mathrm{d}y_{-j}\right) \\ &+ \delta \int_{y_{-j}}\int_{y_j}\frac{\partial V^{comp}(\boldsymbol{\mu'};\theta)}{\partial \mu'_j}\frac{\partial \mu'_j}{\partial \mu_j} f\left(y_j|e_j^\theta,\theta_j\right) f\left(y_{-j}|\boldsymbol{e^*_{-j}}\right)\mathrm{d}y_j\mathrm{d}y_{-j}\end{aligned}$$

Note that the term in parenthesis in the second row is equal to 0 at an optimum. Therefore, we have

$$\frac{\partial V^{comp}(\boldsymbol{\mu};\theta)}{\partial \mu_j} = \frac{\partial \boldsymbol{R}\Big(\boldsymbol{\mu}, \boldsymbol{e}(\boldsymbol{\mu})\Big)}{\partial \mu_j} + \delta\mathbb{E}\left(\frac{\partial V^{comp}(\boldsymbol{\mu'};\theta)}{\partial \mu'_j}\frac{\partial \mu'_j}{\partial \mu_j}\Bigg| e_j^\theta, \theta_j, \boldsymbol{e^*_{-j}}\right) \tag{68}$$

The second term on the right-hand side of expression (68) is "non-standard", and represents the second channel through which current effort decisions generate future benefits. One interpretation of this can be the following: in addition to the direct impact on *current* revenues, a marginal change in the type belief today μ affects the future learning dynamics. This impact is very particular of a Bayesian learning environment. Therefore, such change will affect *all* the future profits for the seller,

[16]Technically, the reason is that, although the seller can only affect the density (i.e. weights) with which the future expected value of reputation is computed, the optimal decision implies shifting probability mass from certain outcomes (y_j) to other outcomes. In other words, the derivative of the density function $f(y|e,\theta)$ with respect to e will be be negative for certain values of y_j and positive for others. This implies that seller j will decide the effort level by "shifting" probability mass such that, marginally, the weighted change in value is the same as the change in the cost of effort.

since all the future generations of buyers will shape their beliefs (and therefore their selection and biding strategies) accordingly[17]. The link with the Euler equation (67) is then clear: an optimal effort strategy (one that maximizes the value for a seller) affects the distribution of beliefs tomorrow, which triggers a change in *expected* short-run nrevenues and a new"learning" path which will affect all future expected profits. The latter impact appears *independently* of the effort decision tomorrow.

E.2 Existence of a Markov Perfect Equilibrium

We focus on Markov Perfect (Bayesian) Equilibibrium concept, as defined in section 3.5. Existence in the dynamic setting is, however, a delicate issue. Given that sellers' actions are continuous, we follow Adlakha et al. (2010) and Escobar (2011) to argue the existence of an equilibrium. Their main results imply the existence of a MPE provided the sets of actions are compact, the set of states is countable, the period payoffs are upper semi-continuous in the action profiles and lower semi-continuous in the competitors' actions and the transition function depends continuously on sellers' actions.

E.3 The (Short-Run) Reputation-Elasticity of Revenues

In our dynamic setting, when deciding about effort levels, sellers care about the marginal effect of reputation on their *lifetime* income from future transactions. As shown in equation (68), the bayesian learning structure implies that, at an equilibrium,

$$\frac{\partial V^{comp}\left(\boldsymbol{\mu};\theta\right)}{\partial \mu_j} \neq \frac{\partial \boldsymbol{R}\Big(\boldsymbol{\mu},\boldsymbol{e}\left(\boldsymbol{\mu}\right)\Big)}{\partial \mu_j} \tag{69}$$

The size of the difference between the left-hand side (LHS) and the right-hand side (RHS) in (69) will be determined by the slope of of the learning function w.r.t μ, that is, $\frac{\partial \mu_j'}{\partial \mu_j}$. This determines how much does a change in current reputation affects the value for a seller due to the impact of the learning structure on future revenues.

[17] Recall that, although buyers are short lived, they observe a summary of past outcomes, represented by μ.

The second term in the envelope condition (68) is an infinite sum of future revenues, weighted by such learning slope and the discount factor. On one hand, when the potential types of sellers are far appart (i.e. $g - b \approx 1$) the learning slope will tend to be relatively big compared to the case when seller types are not very different. On the other hand, however, when sellers are either very trustworthy or not trustworthy, types will be revealed quite fast, meaning that soon into the life of a seller, the reputaion μ will be either 0 or 1 forever.

The equilibrium of the infinite horizon game implies a particular form for the effort strategies; these are a function of current reputation and "deep" parameters, which include the discount factor δ, type values g, b , the number of buyers N in each period, and cost function parameters C. Therefore, we have that equilibrium (expected) revenues at the beginning of a period for seller j can be expressed as

$$\boldsymbol{R_j}\Big(\mu_j, \mu_{-j}, e_j\left(\boldsymbol{\mu};\Omega\right), e_{-j}\left(\boldsymbol{\mu};\Omega\right)\Big) = r_j\Big(\mu_j, e_j\left(\boldsymbol{\mu};\Omega\right)\Big) H_{2j}\Big(\mu_j, \mu_{-j}, e_j\left(\boldsymbol{\mu};\Omega\right), e_{-j}\left(\boldsymbol{\mu};\Omega\right)\Big) \tag{70}$$

where $\Omega \equiv \{\delta, g, b, N, C\}$, $\boldsymbol{\mu} \equiv \{\mu_j, \mu_{-j}\}$, and $e_j\left(\boldsymbol{\mu}\right) \equiv \left\{e_j^g\left(\boldsymbol{\mu}\right), e_j^b\left(\boldsymbol{\mu}\right)\right\}$ are equilibrium effort strategies.

A natural meassure of the impact of reputation is the elasticity of expected revenues in period t for seller j with respect to her reputation in the same period. In equilibrium, such elasticity, which we denote $\eta_{\mu_j}^{R_j}$, depends on effort strategies and is given by (for simplicity, the notation abstract from the dependence on deep parameters):

$$\begin{aligned}\eta_{\mu_j}^{R_j} &\equiv \frac{\partial \boldsymbol{R_j}\Big(\mu_j, \mu_{-j}, e_j\left(\boldsymbol{\mu};\Omega\right), e_{-j}\left(\boldsymbol{\mu};\Omega\right)\Big)}{\partial \mu_j} \times \frac{\mu_j}{\boldsymbol{R_j}\Big(\mu_j, \mu_{-j}, e_j\left(\boldsymbol{\mu};\Omega\right), e_{-j}\left(\boldsymbol{\mu};\Omega\right)\Big)} \qquad (71)\\ &= h\Big(\mu_j, \mu_{-j};\Omega\Big)\end{aligned}$$

The crucial aspect is that $H_{2j}\left(\cdot\right)$ is a function of both sellers effort in equilibrium. This implies that the reputation-elasticity of revenues is potentially a function of *both* reputations, and therefore not constant if reputations evolve over time. In addition to the dependence on the competitor's reputation, and given the characterization results in proposition 5, the dependence onn the number of buyers N, is of interest too. As it has been shown, when the number of buyers in the market increases, their equilibrium behavior implies that sellers startegic incentives disappear. Therefore, we expect the nature of the bias that might arise from reduced form estimations of (71) to change with N too.

E.3.1 Reduced Form Estimation of $\eta_{\mu_j}^{R_j}$?

Recall that, conditional on sellers reputations, there are two sources of uncertainty in the environment we proposed: (i) the valuations that the N buyers have for the good, which are independently drawn each period from $U \sim [0,1]$, and (ii) the selection of auctions characterized in equilibrium by *mixed* strategies. We can write *realized* revenues in equilibrium for seller j at time t as

$$\boldsymbol{RR_{j,t}} = r_j\left(\mu_{j,t}, \mu_{-j,t}\right) v_j^{(2)} \tag{72}$$

where $v_j^{(2)}$ is the *realized* second highest valuation in seller j auction. Importantly, $\boldsymbol{RR_{j,t}}$ and $(\mu_{j,t}, \mu_{-j,t})$ are the variables that a researcher can potentially observe.

Given that we assume sellers form rational expectations, and that conditional on the parameters of the game Ω srategies are a function only of $\boldsymbol{\mu}$, we can think of the expected revenues $\boldsymbol{R_{j,t}}$ as a linear projection of $\boldsymbol{RR_{j,t}}$ on the space of a particular type of functions of $\boldsymbol{\mu_t}$. In other words, we can decompose $\boldsymbol{RR_{j,t}}$ as

$$\boldsymbol{RR_{j,t}} = \boldsymbol{R_{j,t}} + \epsilon_{j,t} \tag{73}$$

where $\boldsymbol{R_{j,t}} = \mathbb{E}\Big(\boldsymbol{RR_{j,t}}\Big|\boldsymbol{\mu_t}\Big)$ and $\epsilon_{j,t}$ is a forecast error with

$$\mathbb{E}\left(\epsilon_j \middle| \boldsymbol{\mu_t}, \boldsymbol{e}\left(\boldsymbol{\mu_t}\right), \boldsymbol{\mu_{t-1}}, \ldots\right) = \mathbb{E}\left(\epsilon_j \middle| \boldsymbol{\mu_t}, \boldsymbol{e}\left(\boldsymbol{\mu_t}\right)\right) = 0 \tag{74}$$

Note that, although the decomposition in (73) arises from a linear projection, $\boldsymbol{R_{j,t}}$ is potentially a non-linear function of $(\mu_{j,t}, \mu_{-j,t})$.

All recent empirical studies about the impact of reputation on revenues, such as Cabral and Hortaçsu (2010) who uses both cross section as well as panel data for sellers, or Resnick et al. (2006) who use field experiment data in order to generate exogenous variation in reputation, estimate a specification similar to the following (log) linear expression for $\boldsymbol{RR_{j,t}}$ in order to try to uncover some information about η_μ^R:

$$\log \boldsymbol{RR}_{j,t} = \beta_0 + \beta_1 \log \mu_{j,t} + X_{j,t}\Gamma + \nu_{j,t} \tag{75}$$

where $\log \boldsymbol{RR}_{j,t}$ are the log-revenues[18] for seller j in period t; $\log \mu_{j,t}$ is the log-public reputation of seller j in period t; X_j is a set of other control variables potentially including seller's fixed characteristics, and $\nu_{j,t}$ is an error term.

Would a simple OLS estimate of β_1, $\hat{\beta}_1^{OLS}$ using either panel or experimental data on revenues and reputation, be enough to consistently estimate η_μ^R? Under general conditions, the answer is likely to be "no". In order to uncover η_μ^R through eq. (75), one needs to make at least two assumptions: (i) the reputation of seller j's competitors $\log \mu_{-j,t}$ does not directly influence seller j's revenues, and (ii) the way in which $\log \mu_{j,t}$ affects $\boldsymbol{RR}_{j,t}$ needs to be constant across μ_j. Under the structure of the simple market presented here, equilibrium revenues depend on effort strategies, which are not observable by the econometrician, and they are a function of *both* sellers reputations. In other words,

$$\nu_{j,t} = g\Big(, e_j\left(\boldsymbol{\mu}\right), e_{-j}\left(\boldsymbol{\mu}\right); \epsilon_{j,t}\Big) \tag{76}$$

In equilibrium, it will probably be the case that $Cov\Big(\log \mu_{j,t}, \nu_{j,t}\Big) \neq 0$: not only might $\mu_{j,t}$ and $\mu_{-j,t}$ be correlated through past and current effort, but the dependence of RR on $\mu_{j,t}$ is potentially not constant. The latter implies that even an exogenous source of variation in $\mu_{j,t}$, say from a field experiment as in Resnick et al. (2006), would not be enough to uncover most of the information about the elasticity.[19]

[18] Due to lack of time series data on past prices, Cabral and Hortaçsu (2010) proxy revenues by the number of sales. For our purpose, however, they both measure similar aspects of trades.

[19] Resnick et al. (2006) generate such exogenous variation through an experiment in which a seller with high reputation creates an additional account under a different name. Under this new account, the auction characteristics are the same as the ones in the auctions under her original name, but reputation would be clearly different. The relevant parameter, which in essence is their counterpart of β can be computed as

$$\tilde{\beta} = \bar{\boldsymbol{RR}}_j^{original} - \bar{\boldsymbol{RR}}_j^{new}$$

The next relevant question is then: can a reduced form estimation such as (75) provide some useful information about η_{μ}^{R}, *even* when η_{μ}^{R} cannot be consistently identified by $\hat{\beta}_1^{OLS}$? Consider the unweighted *average* of $\eta_{\mu_j}^{R_j}$ across the whole state space (μ_j, μ_{-j})

$$\bar{\eta}_j = \int_{M_j} \int_{M_{-j}} h\Big(\mu_j, \mu_{-j}; \Omega\Big)\, d\,\mu_{-j}\, d\,\mu_j \tag{77}$$

E.4 Quantitative Experiment

In this section we quantify importance of the endogeneity problem described above, which can arise when strategic interactions and reputation building behavior are not taken into account in the reduced form estimation.

We first propose an algorithm to solve the infinite-horizon game, and present the quantitative properties of the approximated equilibrium (including the true elasticity of revenues) for different parameter values. We then calibrate a benchmark version using moments presented in Cabral and Hortaçsu (2010), and use simulated data to run different regressions that allow us to analyze quantitatively the importance of omitting reputation building and strategic behavior from the reduced form model.

E.4.1 Computational Algorithm

In order to solve the dynamic game for a given set of parameter values, we propose an algorithm in the spirit of Doraszelski and Pakes (2007) solving for the value functions of each seller and type, while searching for an equilibrium using the contraction properties of best responses. Some important aspects that the algorithm needs to take into account:

There are four key features of the dynamic game that must be taken into account when solving the problem numerically:

1. Since effort decisions are not observable, they do not affect the type-beliefs directly, but indirectly through the information (signal) revealed by the outcomes y_j. As described in the previous section, an effort decision $e_{j,t}$ at time t (i.e.

.

conditional on the information set at time t) will have an impact on all the probabilities of future outcome sequences. This feature prevents us from using a numerical algorithm that works only through the Euler equations since these would be composed of an infinite sum of terms (i.e. future profits) weighted by the change in probabilities after an effort decision at time t.

2. Probabilities of outcomes (deliveries) and beliefs are both *subjective perceptions of seller j*, function of the best responses (BR) for all players, and these perceptions must be confirmed in equilibrium. From seller j's point of view:

$$P\left(y'_j, y'_{-j}|\theta_j, BR_j\left(\boldsymbol{e^*_{-j}}, \boldsymbol{s}\right)\right) = P\left(y'_j|\theta_j, e^\theta_j\right) \cdot P\left(y'_{-j}|\boldsymbol{e^*_{-j}}, \mu_{-j}\right) \tag{78}$$

where the equality in (78) comes from the independence of conditional outcomes. Note that $P\left(y'_{-j}|\boldsymbol{e^*_{-j}}, \mu_{-j}\right)$ is the *reputation* of seller $-j$, which represents a combination of *belief* and perceptions held by buyers (and therefore relevant for other sellers).

3. The dynamic reputation game is characterized by a *continuous action space* (the effort chosen by each seller-type is a continuous variable) and a continuous state space (the beliefs about a seller being of good type are also continuous in $[0, 1]$). As it is well known in the game theoretic literature, the existence and uniqueness of an equilibrium under such features is quite different than under a *finite* action space (see for example Athey (2001)). Therefore, the approximated solution needs to preserve such continuity in order to avoid one possible convergence problem. Within each iteration over $\boldsymbol{e^*}$, a new set of value functions (one for each seller and type) will be found. We approximate these (unknown) functions by means of linear approximation schemes, consisting of *basis* functions and approximation *nodes*; in other words, we will have $V_j(\boldsymbol{\mu};\theta) \approx \hat{V}_j(\boldsymbol{\mu}; \theta, \{c\})$ where $\{c\}$ is a vector of parameters.

The algorithm employed is a slightly modified version of the ideas developed in Ericson and Pakes (1995) and Doraszelski and Pakes (2007). We use a *successive* *approximation* strategy to the value functions of all players and types. Intuitively, we want to find (one of the possible) equilibrium in which the perceptions about effort, namely e^*, are indeed optimal strategies by sellers. The algorithm will find a *fixed point* of a mapping

$$\tilde{e}_j^\theta = F\left(\boldsymbol{e^*}, \cdot\right) \quad j = 1, 2\ ;\ \theta = g, b$$

where $\boldsymbol{e^*} = \left\{e_j^{g*}, e_j^{b*}\right\}_{j=1,2}$ and $\tilde{e}_j^\theta$ is the optimal effort strategy for seller j and type θ.

The numerical algorithm used can be summarized as follow:

- *Step 0*: Choose an approximating method, m, and an objective function for the approximation. Here we use one spectral method (Chebyshev polynomials) and one finite-elements method (bi linear splines). The grid G_m (method m, with $m = \{cheb., spline\}$) used to approximate e_j^θ will be defined accordingly.

- *Step 1*: **Perceived effort iteration (PEI).** Choose an initial value for the perceived effort strategies, $e_{j,m}^{0*}\left(\mu_1, \mu_2; \theta\right),\ j = 1, 2; \theta = \{g, b\}$ defined at the grid G_m for the approximating method m. Define a stopping criterion $\epsilon^{PEI} > 0$.

- *Step 2*: Given $e_{j,m}^{0*}$, and all the possible states (beliefs) today (μ_1, μ_2) compute the possible values of the beliefs tomorrow, (μ_1', μ_2'), which depend only on the realizations of the transactions, y_1, y_2

  $$\mu_j' = g\left(\mu_j, e_{j,m}^{0*}, y_j\right),\ j = 1, 2\ ; \theta \in \{g, b\}$$

  where the law of motion $\mu' = g\left(\mu, \cdot\right)$ is derived from a Bayesian learning process.

- *Step 3*: **Value function iteration (VFI)**. Guess an initial value function $V_j^0\left(\boldsymbol{\mu};\theta\right)$ on the grid G_m for each seller j and type θ. Define a stopping criterion $\epsilon^{VFI} > 0$.

- *Step 4*: Projection step[20]. Use such guess and $\boldsymbol{\mu'}$ computed in *step 2* in order

[20] For the case of Chebyshev polynomials, we need to define the *number* of basis elements $\{T\}$, n_1 and n_2 (degree of approximating polynomials for both dimensions). Let $T_{i1}\left(h\left(\mu_1\right)\right)$ and $T_{j2}\left(h\left(\mu_2\right)\right)$

to approximate $V_j^0(\boldsymbol{\mu'};\theta)$ as

$$\hat{V}_j(\boldsymbol{\mu'};\theta,\{\boldsymbol{c_{ij}}\}) \equiv \sum_{k=0}^{n_1}\sum_{l=0}^{n_2} c_{kl}\cdot T_{i1}(\mu'_1)\,T_{j2}(\mu'_2)$$

where the coefficients $\{\boldsymbol{c_{kl}}\}$ are defined as

$$\{c\} = arg\min \int_{\mu_1\times\mu_2}\left[V_j^0(\boldsymbol{\mu};\theta) - \hat{V}_j(\boldsymbol{\mu};\theta\{c\})\right]^2 d\mu_1 d\mu_2 \tag{79}$$

- *Step 5*: Maximization step. Compute $V_j^1(\cdot)$ by solving the linear programming problem

$$\begin{aligned} V_j^1(\mu_j,\mu_{-j};\theta) \;=\; & \max_{e_j^\theta}\Bigg\{\boldsymbol{R}(\mu_j,\mu_{-j},\boldsymbol{e_m^{0*}}) - c\left(e_j^\theta\right) + \\ & +\delta\sum_{y_j}\sum_{y_{-j}} p\left(y_j,y_{-j}|e_j^\theta,\boldsymbol{e_m^{0*}},\mu_{-j}\right)\hat{V}_j\left(\mu'_j(y_j),\mu'_{-j}(y_{-j});\theta\right)\Bigg\} \\ s.t.\;\; & e_j^\theta\in[\underline{e},\bar{e}] \end{aligned} \tag{80}$$

- *Step 6*: If $\max\left\{|V_j^1 - V_{j,}^0|\right\}_{j,\theta} < \epsilon^{VFI}$, stop and go to step 7. Else, update the guess by weighted function iteration: $V_j^0 = \alpha V_j^1 + (1-\alpha)V_j^0$, with $\alpha\in[0,1]$ and go to step 4.

- *Step 7*: If $\max\left\{|\hat{e}_{j,m}^\theta - e_{j,m}^{0*}|\right\}_{j,\theta} < \epsilon^{PEI}$, where $\hat{e}_{j,m}^\theta = arg\max_{e_j^\theta}$ (80), stop. Else, update the guess by weighted function iteration: $e_{j,m}^{0*} = \alpha\hat{e}_{j,m}^\theta + (1-\alpha)e_{j,m}^{0*}$, with $\alpha\in[0,1]$ and go to step 2.

be the i-th (j-th) element of the n_1-th (n_2-th) degree Cheb. polynomial, where $h(\cdot):[0,1]\to[-1,1]$. Given the properties that characterize the Chebyshev polynomial family, the coefficients $\{c_{ij}\}$ are relatively easy to compute (this includes the approximation of the integral in (79) using Gauss-Chebyshev quadrature, where we need to define the number of Chebyshev nodes $m_{\mu 1}$ and $m_{\mu 2}$, with $n_1\le m_{\mu 1}$ and $n_2 < m_{\mu 2}$) .
For the case of bi-linear splines, we choose a set of nodes (in the space of beliefs) $\{\mu_{1i},\mu_{2j}\}_{i,j}$ which together with the basis $\{\psi\}_{i,j}$, they define the tensor approximation as

$$\hat{e}_j^\theta(\mu_1,\mu_2;\{c\}) \equiv \hat{e}(\mu_1,\mu_2,\theta;\{c\}) = \sum_{i=0}^{m_{\mu 1}}\sum_{j=0}^{m_{\mu 2}} c_{ij}\cdot\psi_i(\mu_1)\,\psi_j(\mu_2)$$

The basis (*tent* functions) are such that the coefficients take a simple form: $c_{ij} = V^\theta(\mu_{1i},\mu_{2j})$. In addition, for $\mu_1=\mu_{1i}$ and $\mu_2=\mu_{2j}$ we have $\hat{e}^\theta(\mu_{1i},\mu_{2j}) = e^\theta(\mu_{1i},\mu_{2j})$ Note that, contrary to the strategy folowed for Chebyshev polynomials, we will have a *just-identified* system.

Some particularities of our model are worth noting. The fact that the stochastic vector of outcomes in our model, $\boldsymbol{y}$, (the outcome of transactions given effort decisions and types) takes discrete values (success or failure), implies that we don't need to employ numerical integration when computing expectations in *step 5*; this eliminates one source of numerical approximation error. However, we are still left with the errors arising from the approximation of $V_j(\cdot)$ in *step 4* (*fitting* step); in addition, we cannot avoid incurring in errors coming from an optimization or internal root finding algorithm required in *step 5*.

Given the assumptions regarding the strategies we allow for each seller, the numerical solution provided (time-invariant) effort decision rules.[21] Accuracy tests for the approximated solution and simulation are presented in the appendix.

E.4.2 Parametrization and Calibration

[TBW]

[21] To be precise, what we describe here is an "approximate" equilibria. As noted by **?**, such equilibriacan be far away from the "exact" equilibria. They provide sufficient conditions in order to ensure that such approximated equilibria is close to the exact one. Although such conditions are presented for competitive economies with heterogeneous agents, we suspect that similar concerns might arise in our dynamic game.